

Analisis de Elasticidad de Precios

OfficeMax | Base 19 mayo 2026 | TEC de Monterrey

PARTE 1

$$\log(\text{qty}+1) = \alpha + \beta \cdot \log(\text{precio})$$

PARTE 2 (multivariable)

$$\begin{aligned} \log(\text{qty}+1) = & \alpha + \beta \cdot \log(\text{precio}) \\ & + \delta \cdot \text{tienda} + \omega \cdot \text{mes} + \gamma \cdot \text{marca} \\ & + \varepsilon \cdot \text{promo} \end{aligned}$$

4

20

49

-0.112

Categorías

SKUs analizados

Estim. P1 sig.

Beta global P2



Elastico ($\beta < -1$)

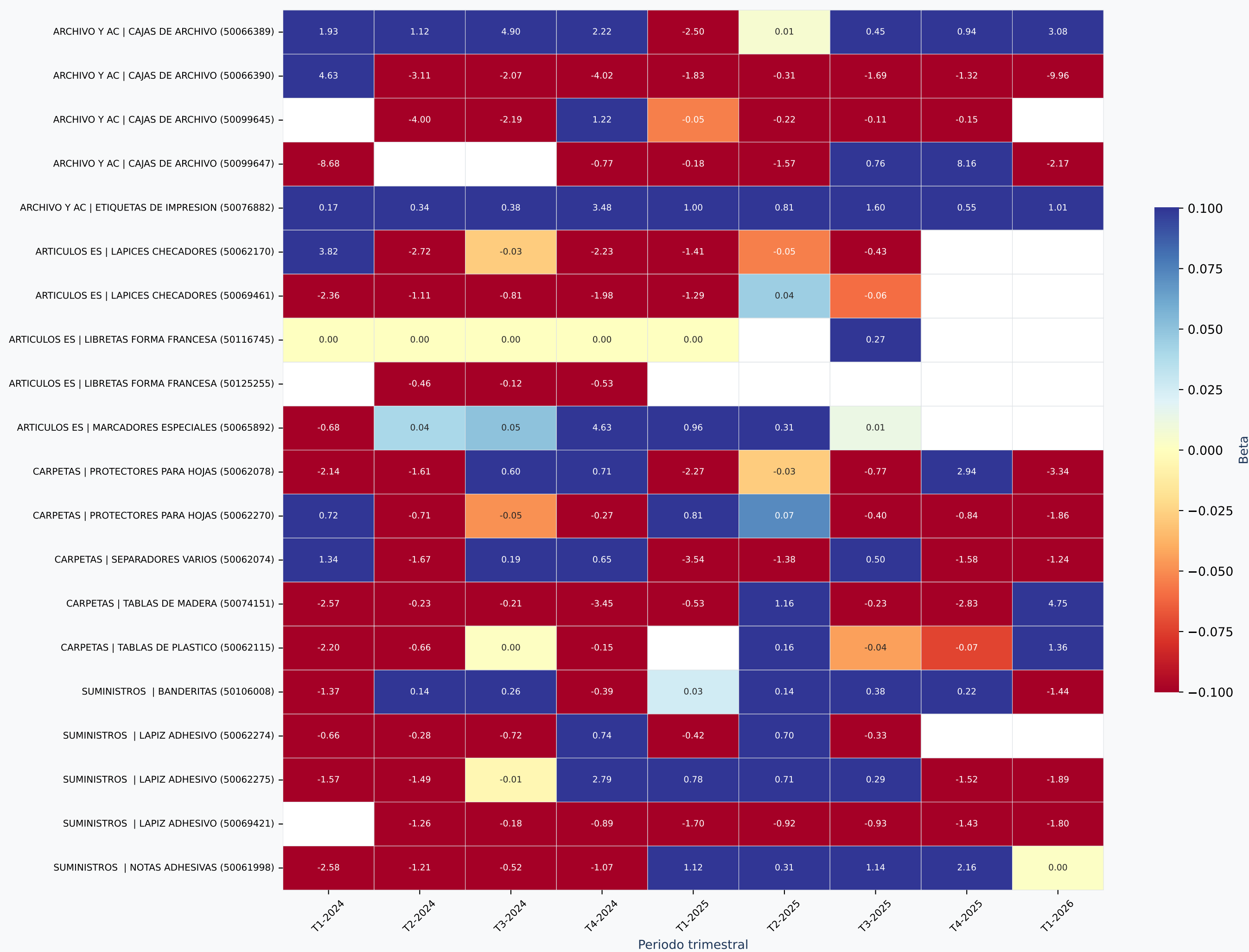


Inelastico ($-1 < \beta < 0$)

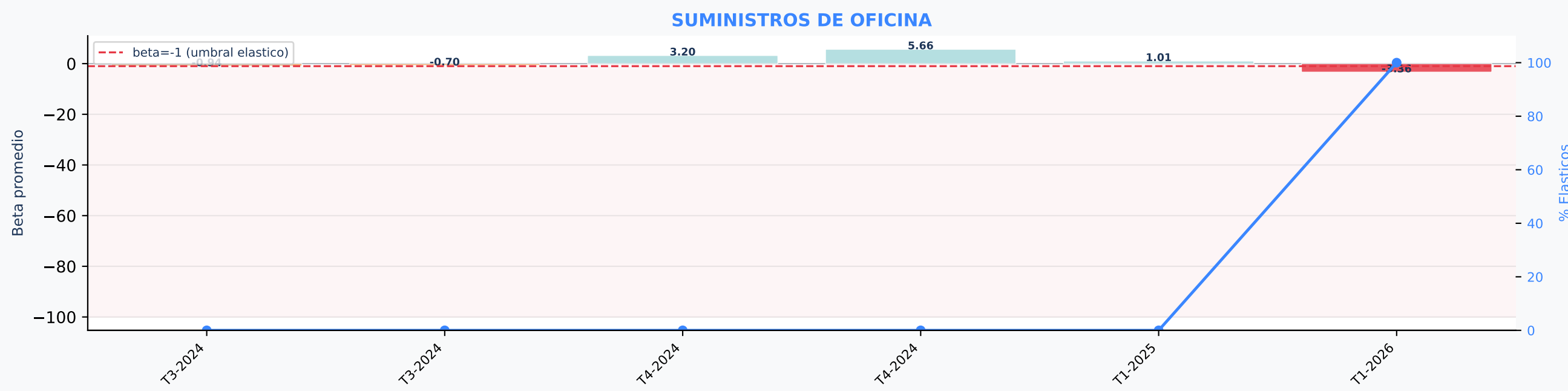
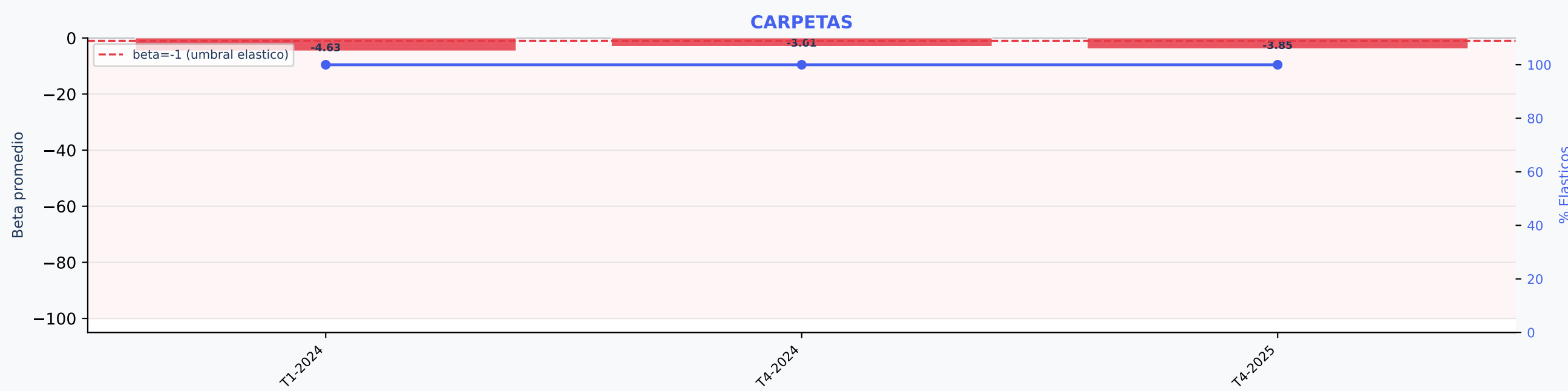
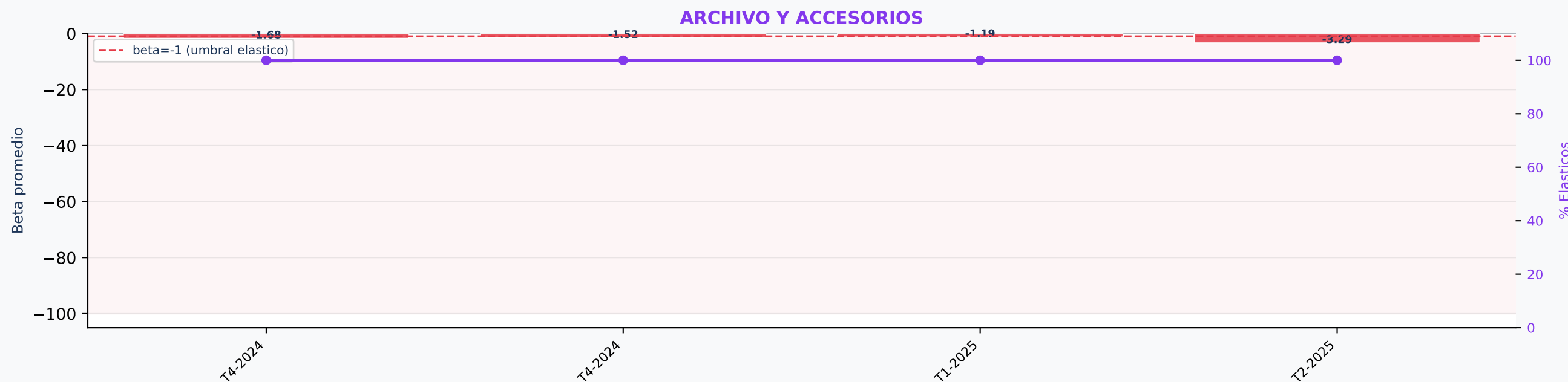


Anomalo ($\beta \geq 0$)

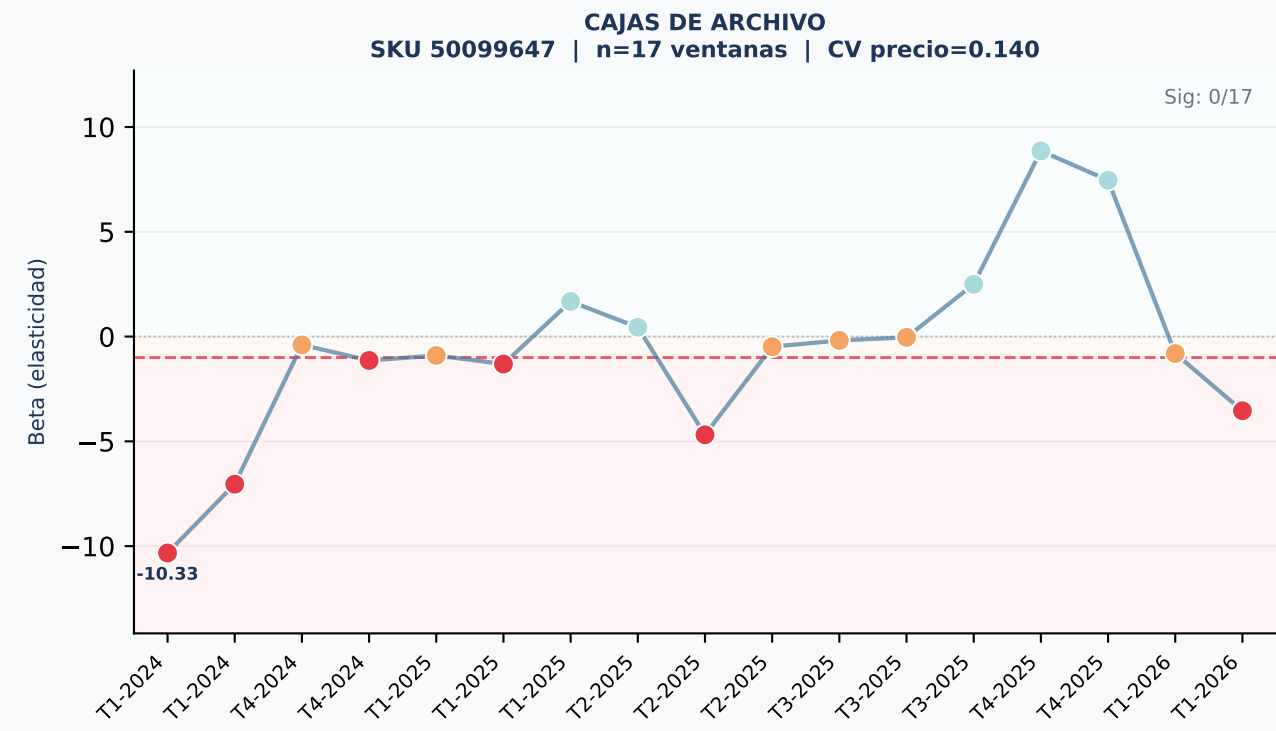
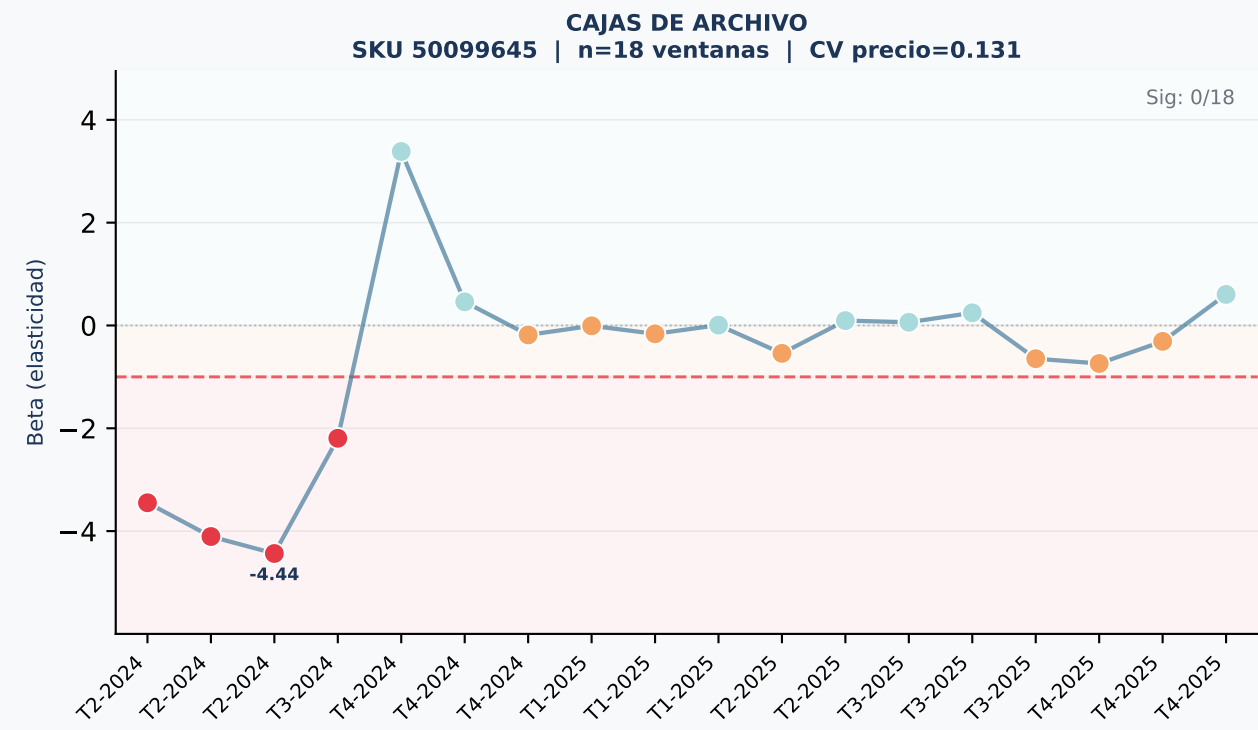
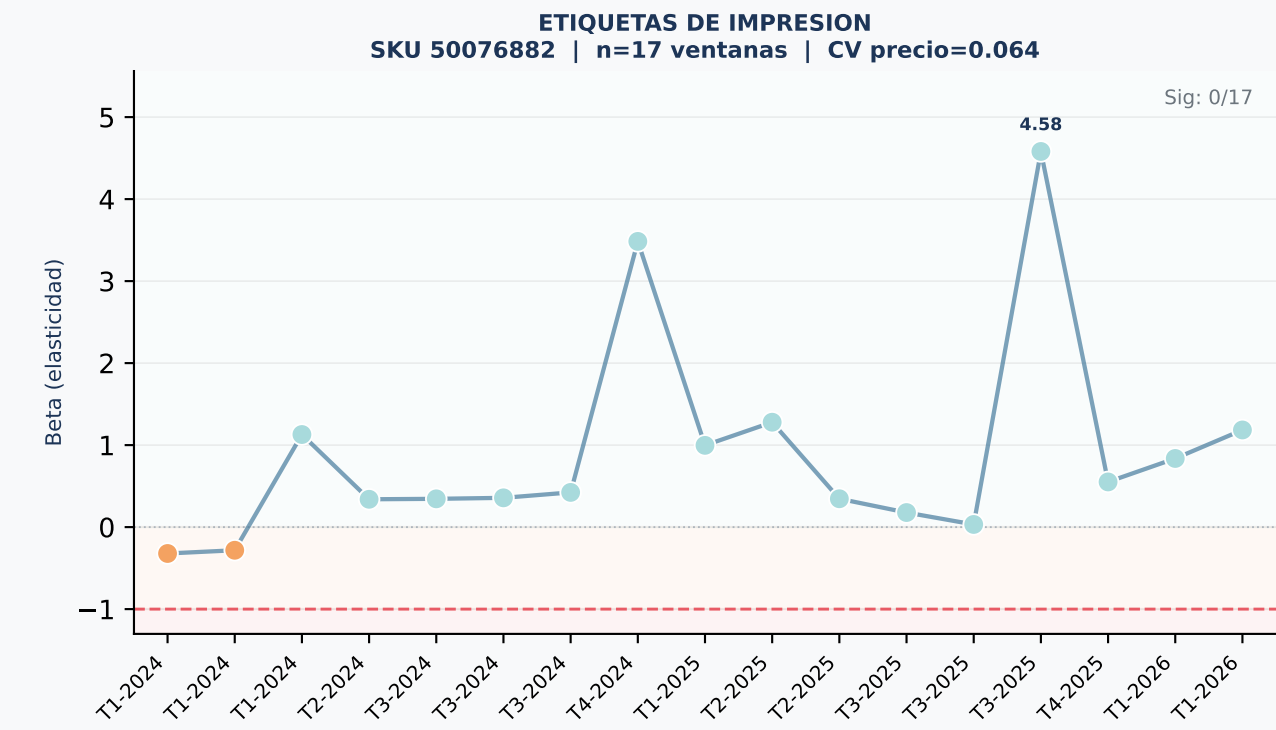
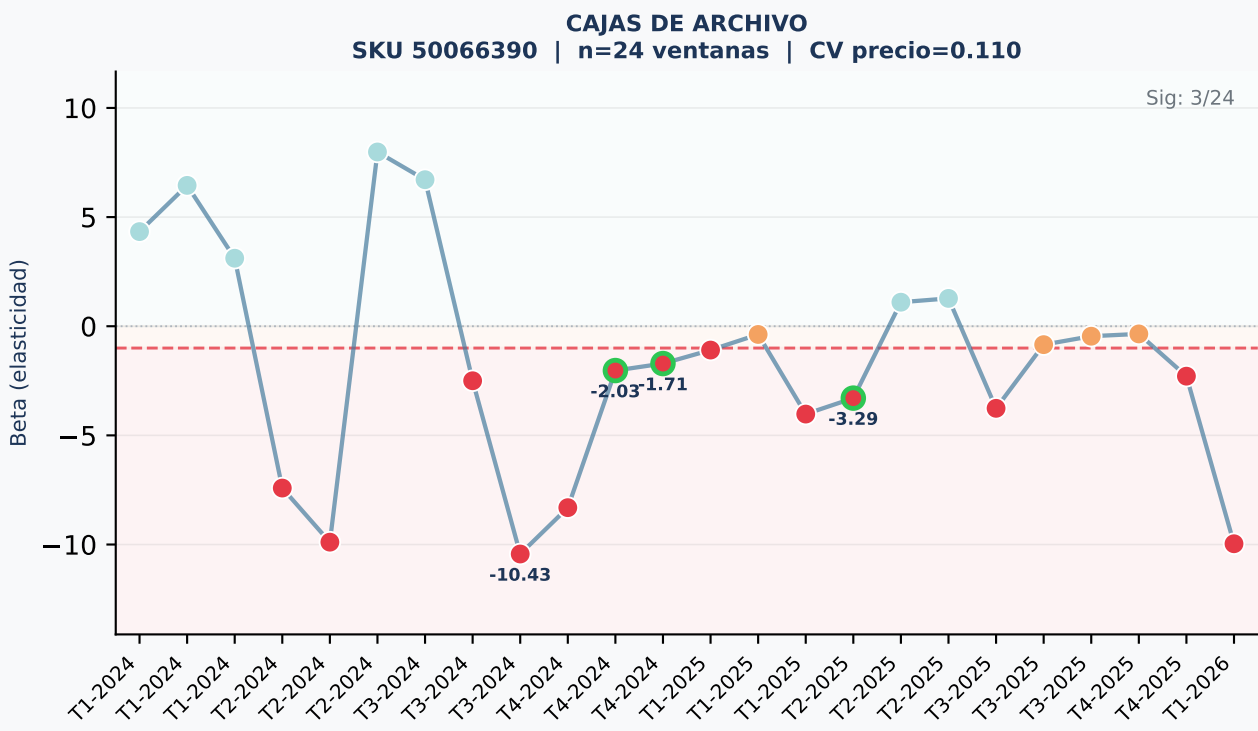
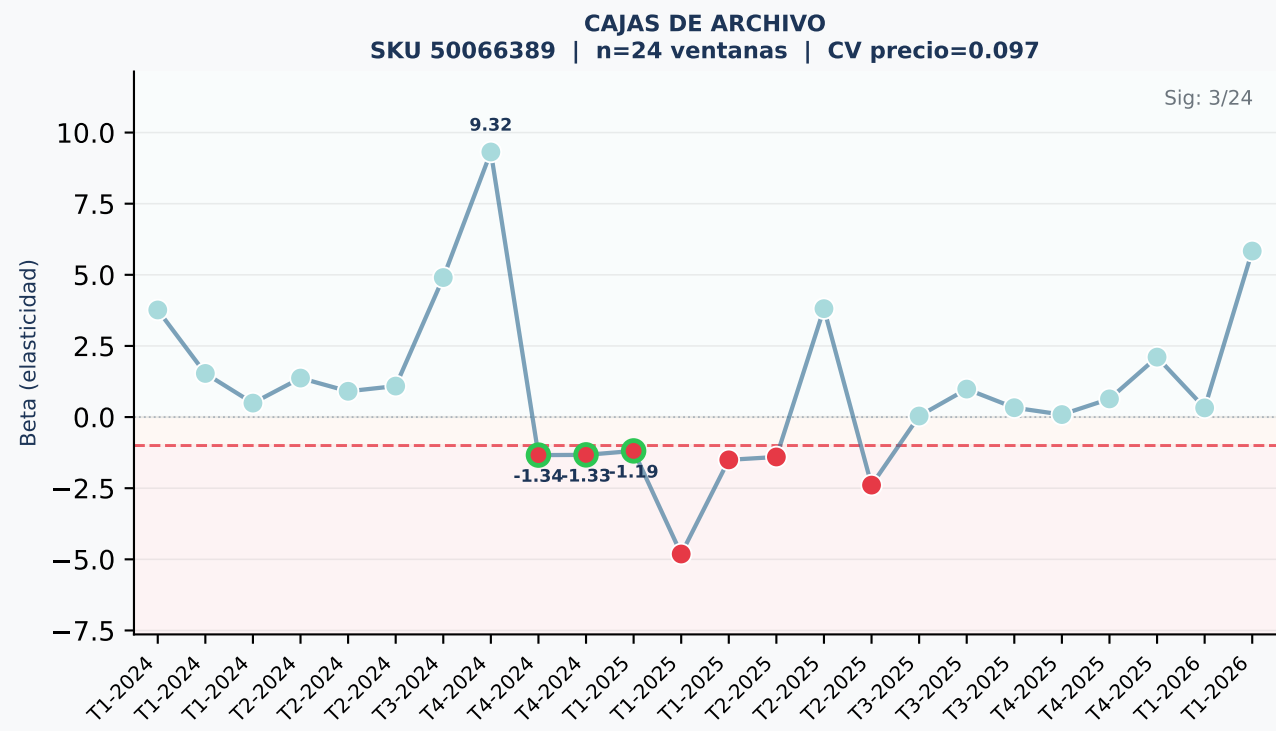
Mapa de Beta Trimestral por SKU (todas las ventanas, sig. y no sig.)
rojo = demanda cae con precio (elastico) | azul = demanda sube con precio (anomalo)



Estacionalidad Trimestral por Categoría

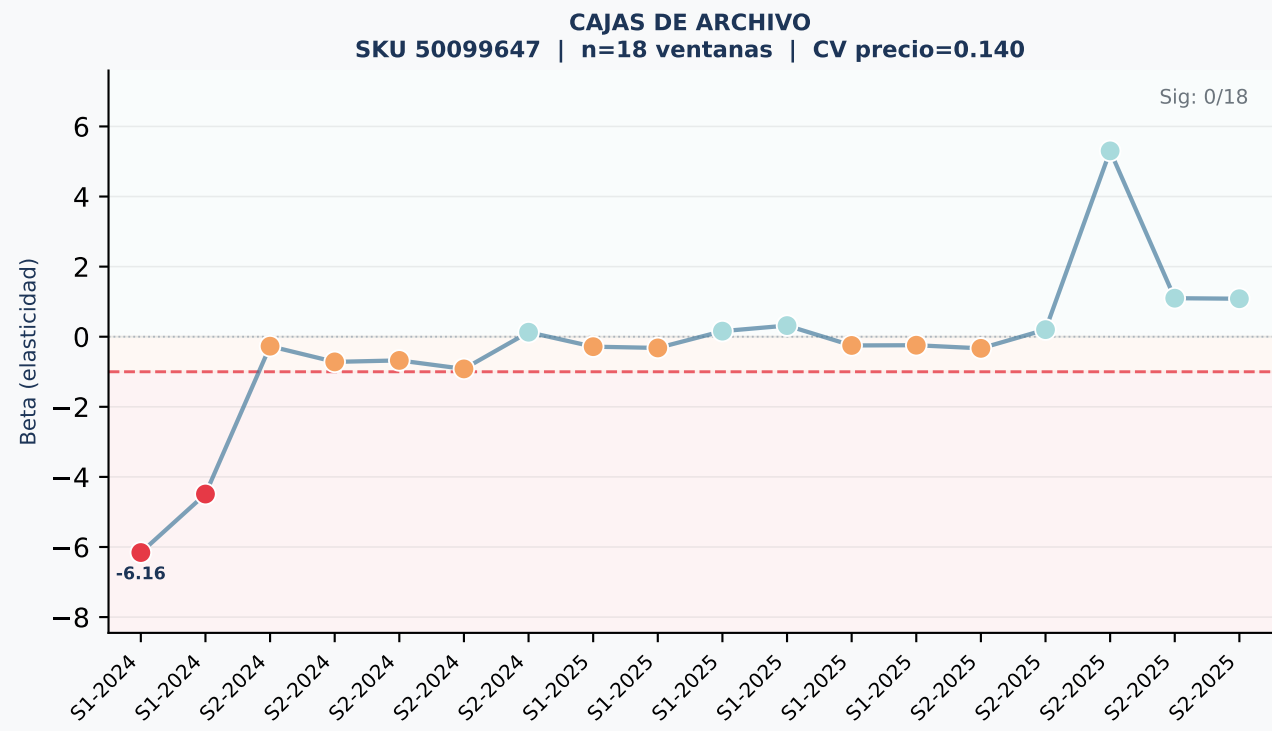
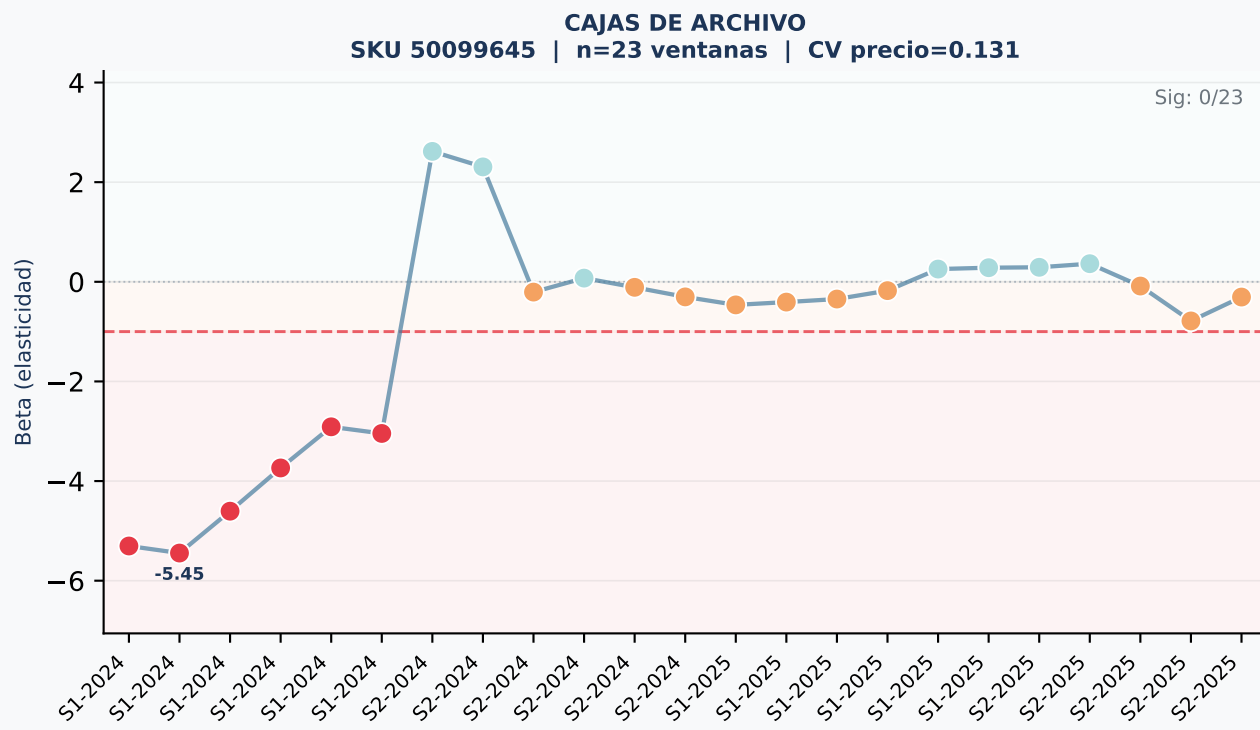
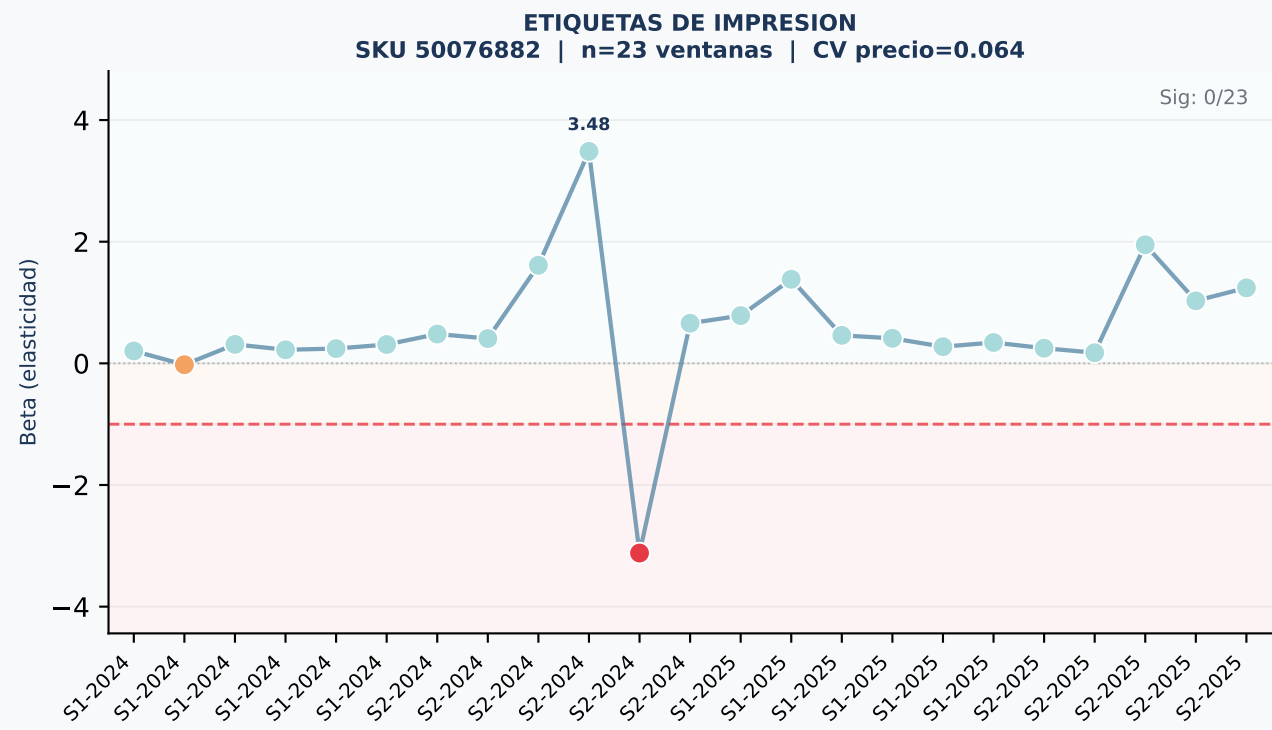
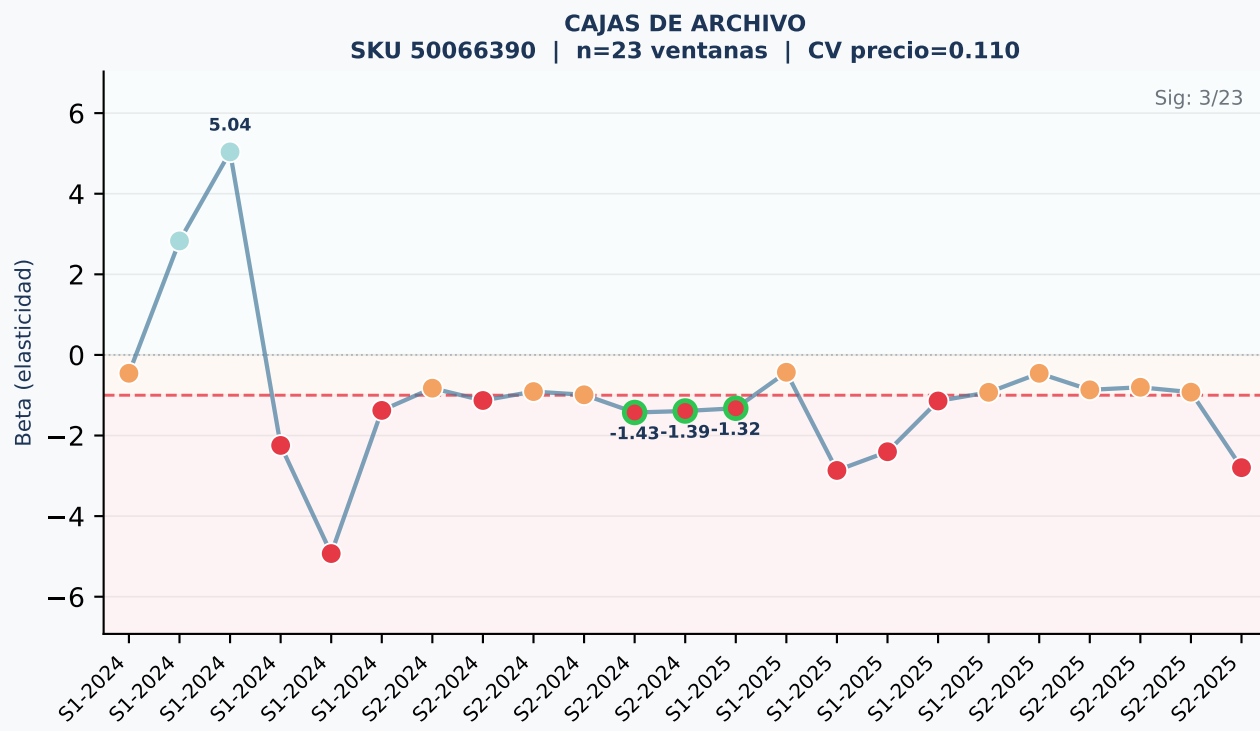
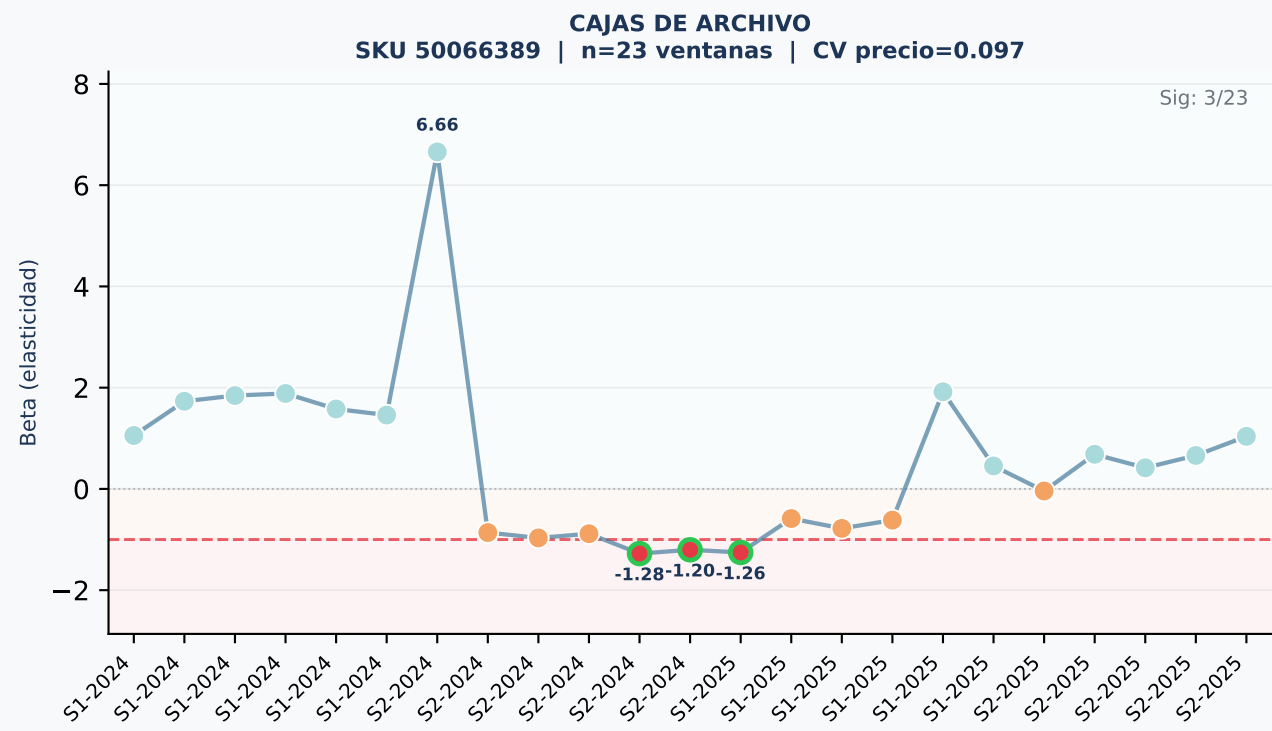


ARCHIVO Y ACCESORIOS — Evolucion de Beta (Trimestral)
Cada punto = ventana de 3 mes(es). Verde = significativo (p<0.05)



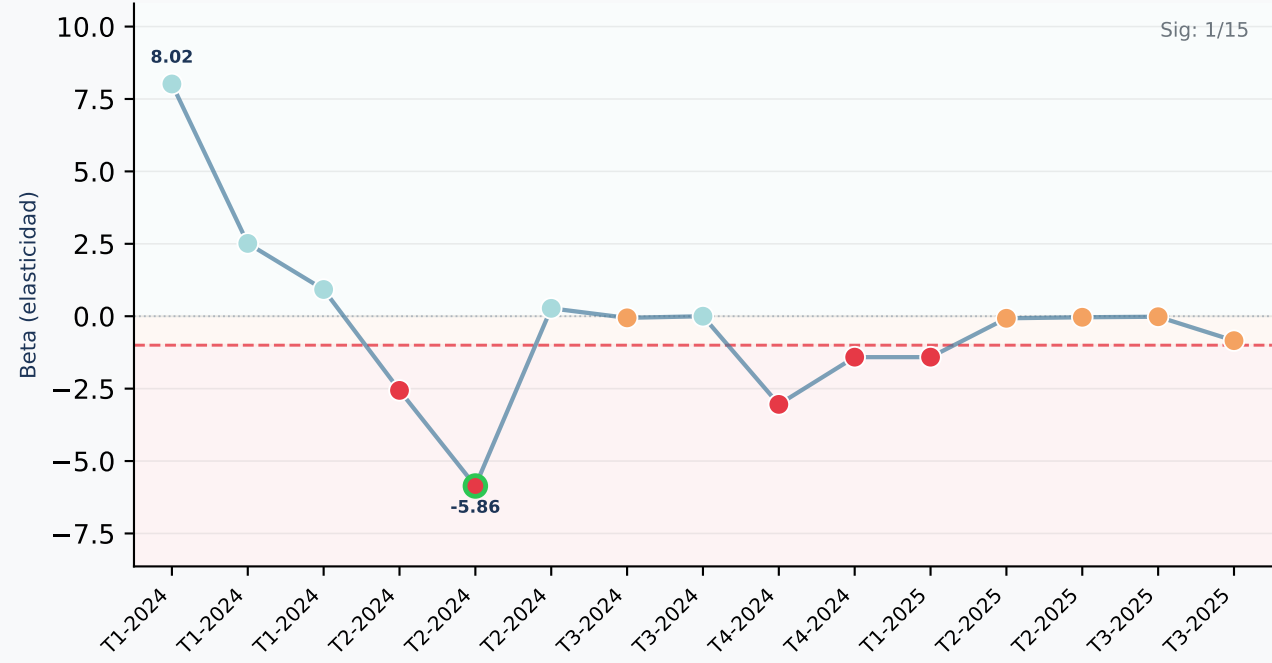
Elastico ($\beta < -1$) Inelastico ($-1 < \beta < 0$) Anomalo ($\beta \geq 0$) Punto significativo ($p < 0.05$)

ARCHIVO Y ACCESORIOS — Evolucion de Beta (Semestral)
Cada punto = ventana de 6 mes(es). Verde = significativo (p<0.05)

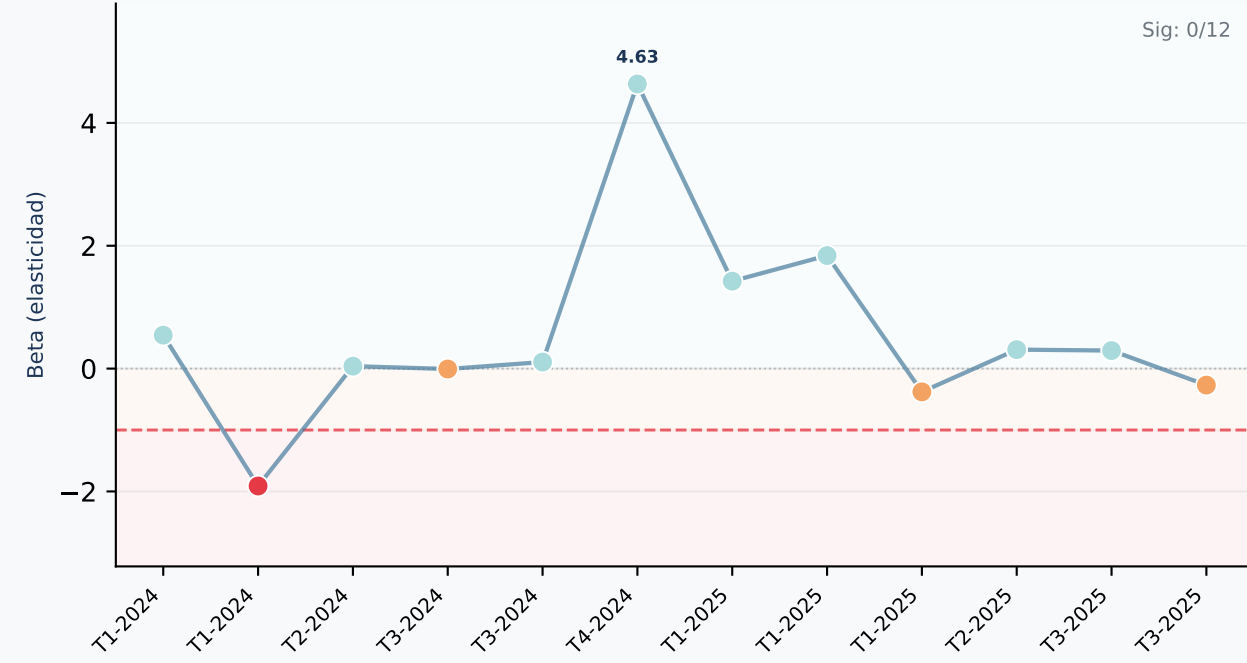


ARTICULOS ESCOLARES — Evolucion de Beta (Trimestral)
Cada punto = ventana de 3 mes(es). Verde = significativo (p<0.05)

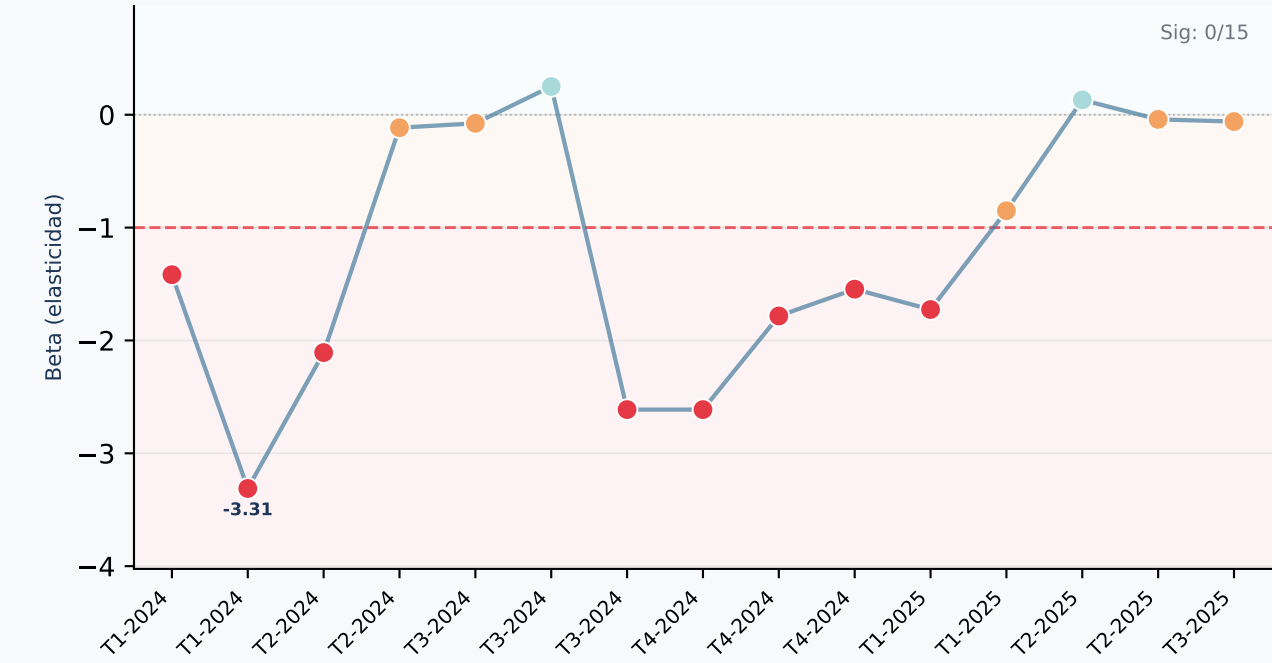
LAPICES CHECADORES
SKU 50062170 | n=15 ventanas | CV precio=0.136



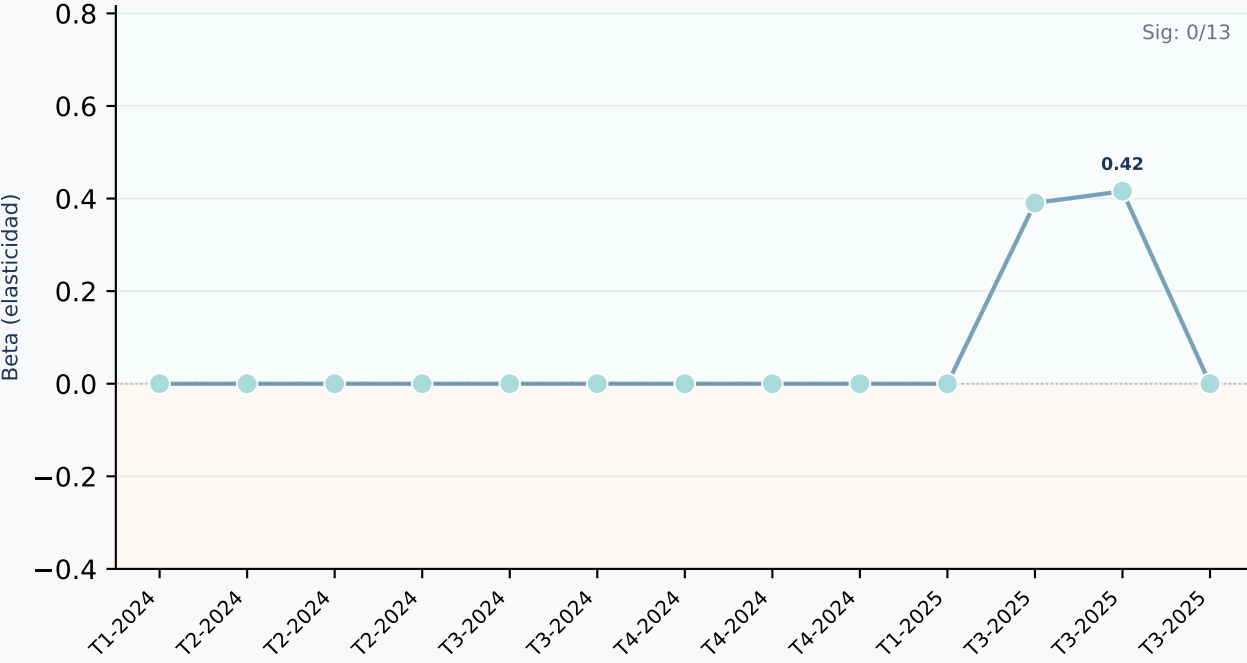
MARCADORES ESPECIALES
SKU 50065892 | n=12 ventanas | CV precio=0.082



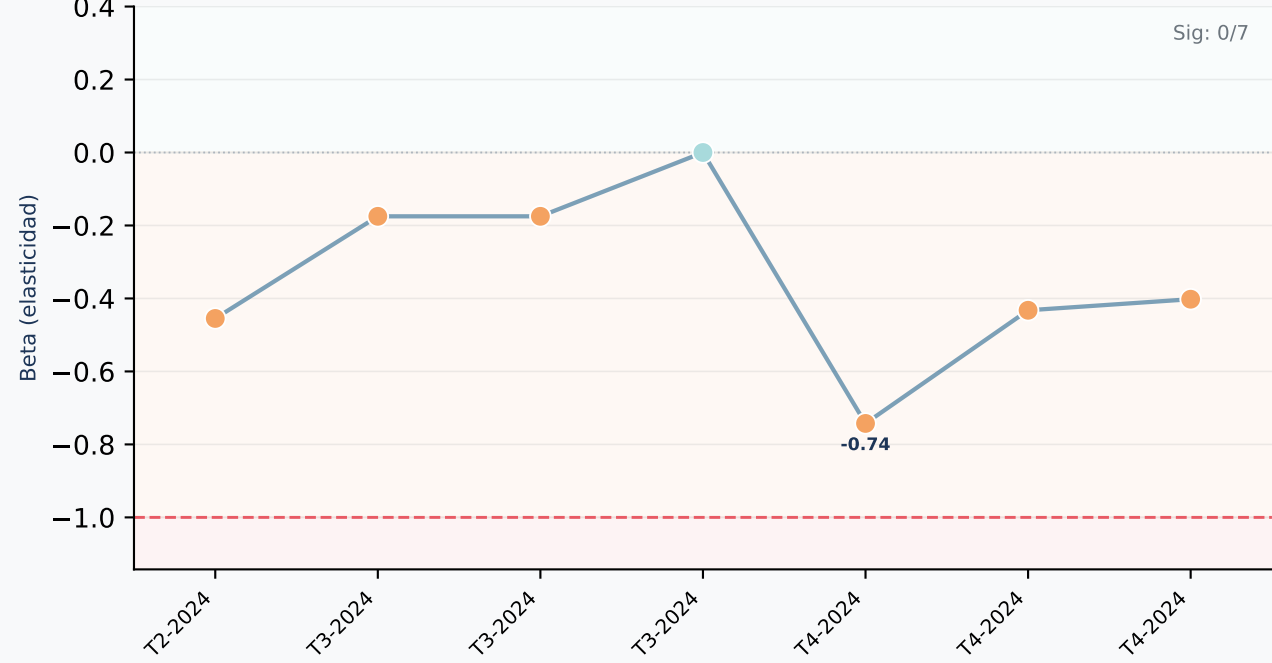
LAPICES CHECADORES
SKU 50069461 | n=15 ventanas | CV precio=0.103



LIBRETAS FORMA FRANCESA
SKU 50116745 | n=13 ventanas | CV precio=0.122

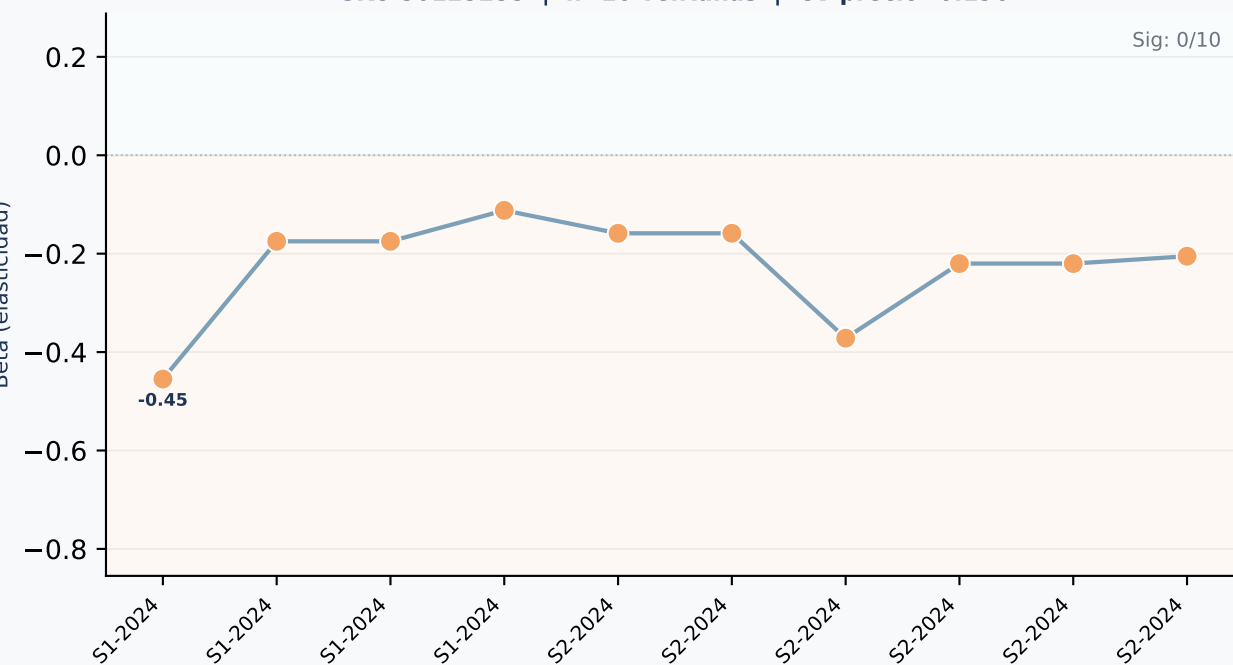
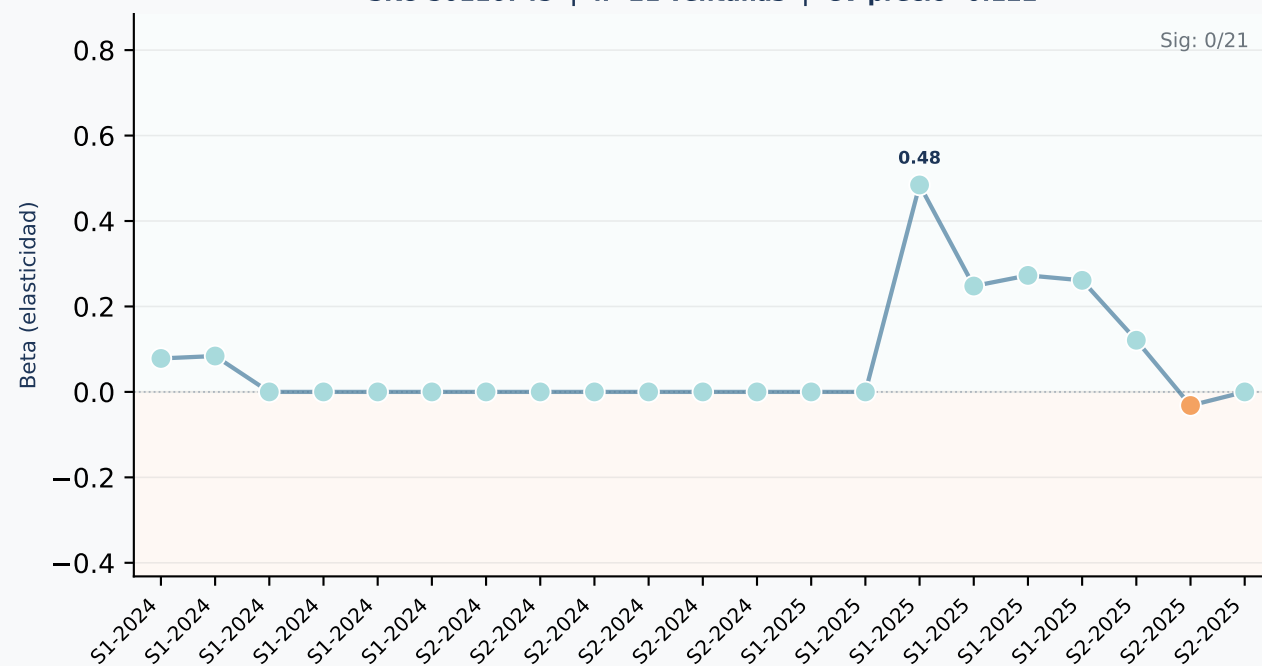
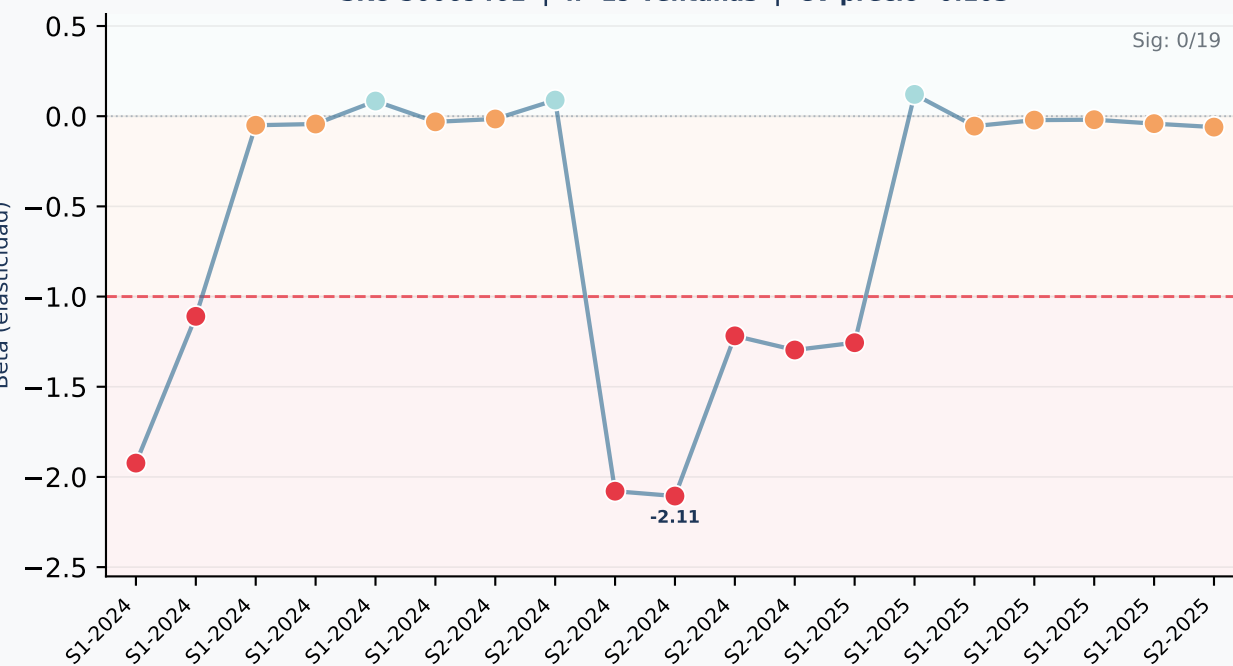
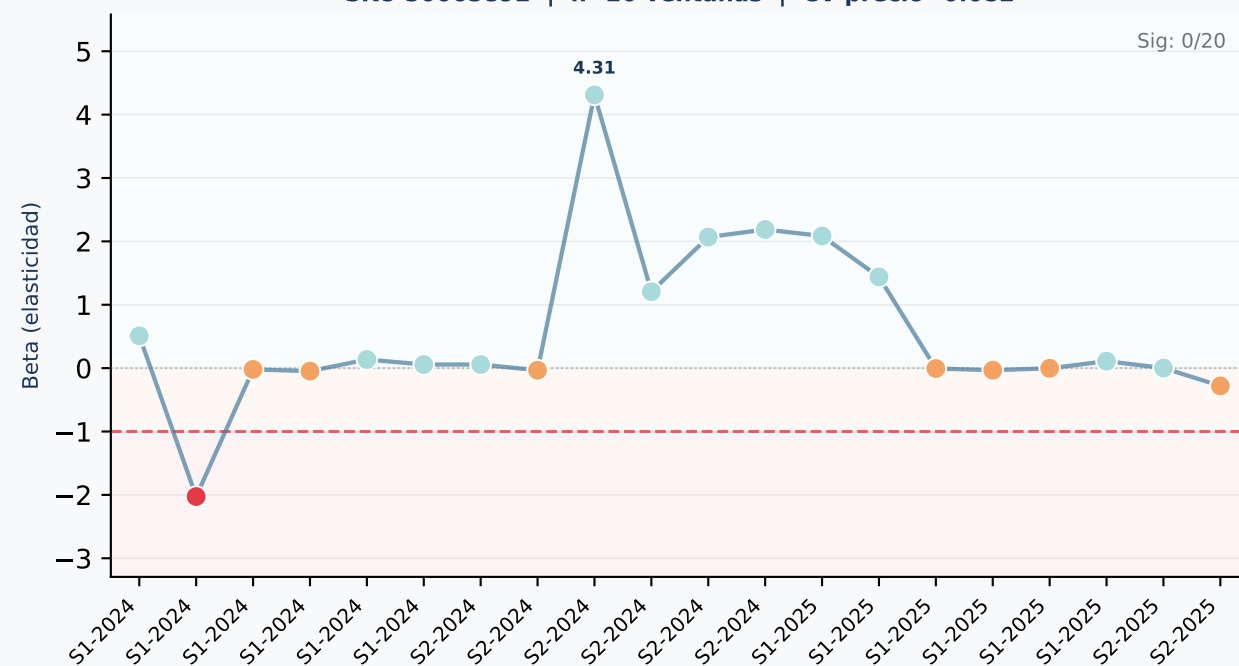
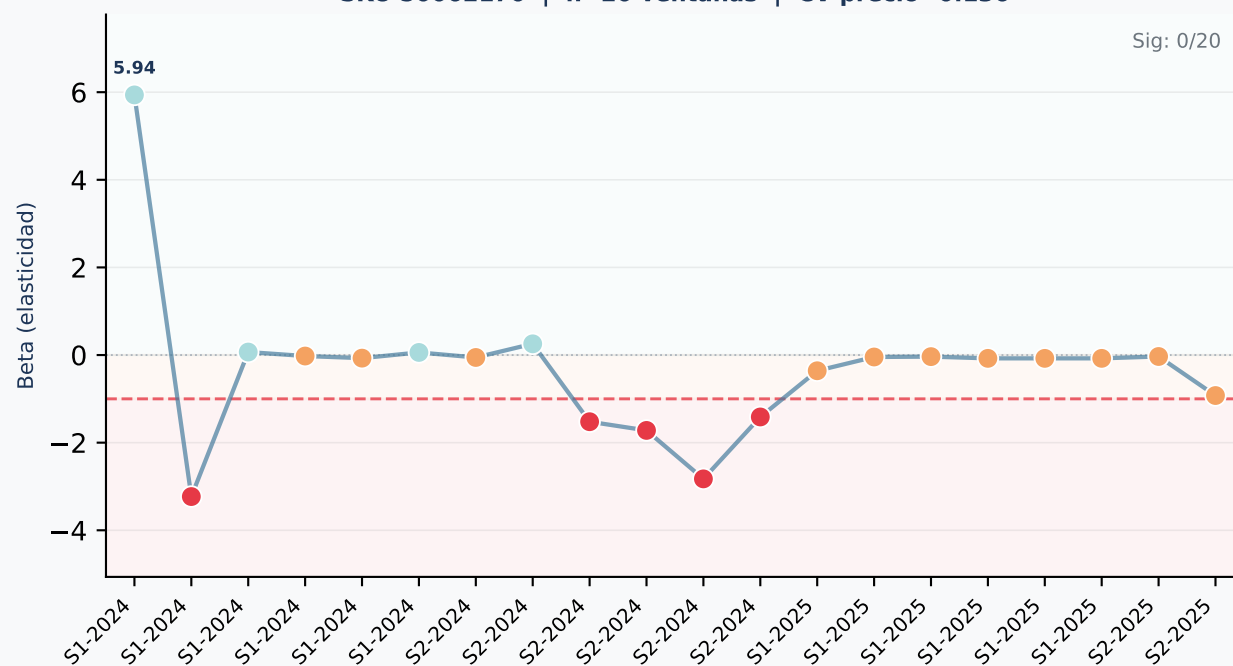


LIBRETAS FORMA FRANCESA
SKU 50125255 | n=7 ventanas | CV precio=0.196



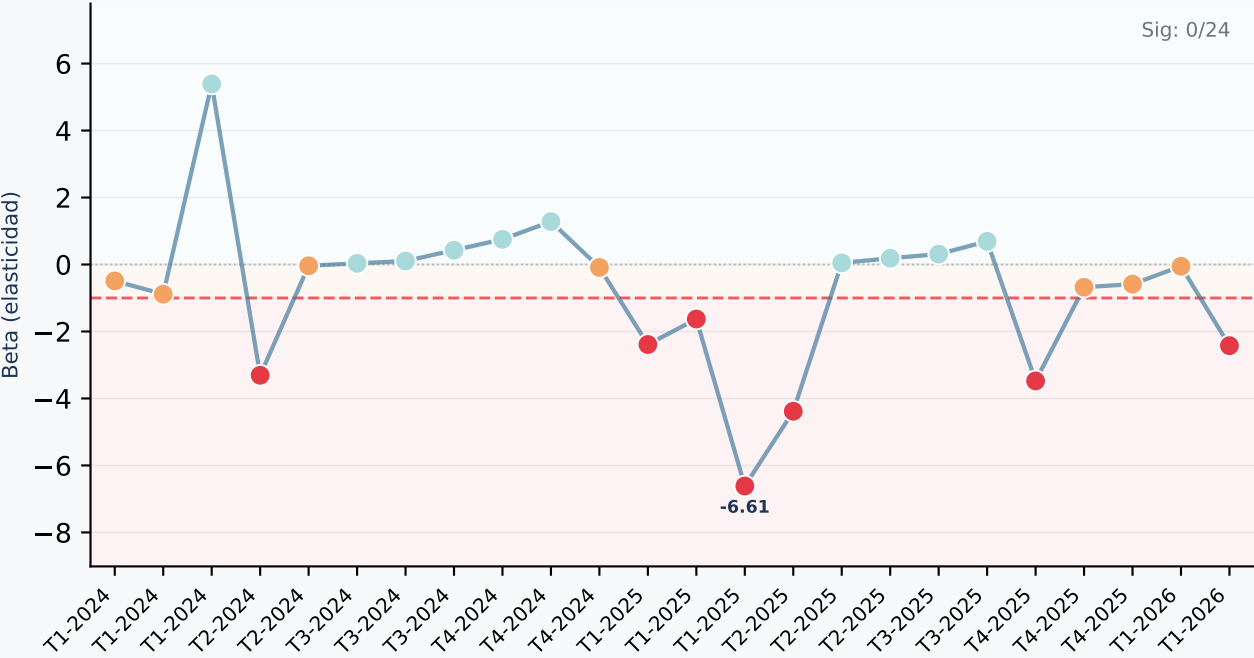
Elastico ($\beta < -1$) Inelastico ($-1 < \beta < 0$) Anomalo ($\beta \geq 0$) Punto significativo (p<0.05)

ARTICULOS ESCOLARES — Evolucion de Beta (Semestral)
Cada punto = ventana de 6 mes(es). Verde = significativo ($p < 0.05$)

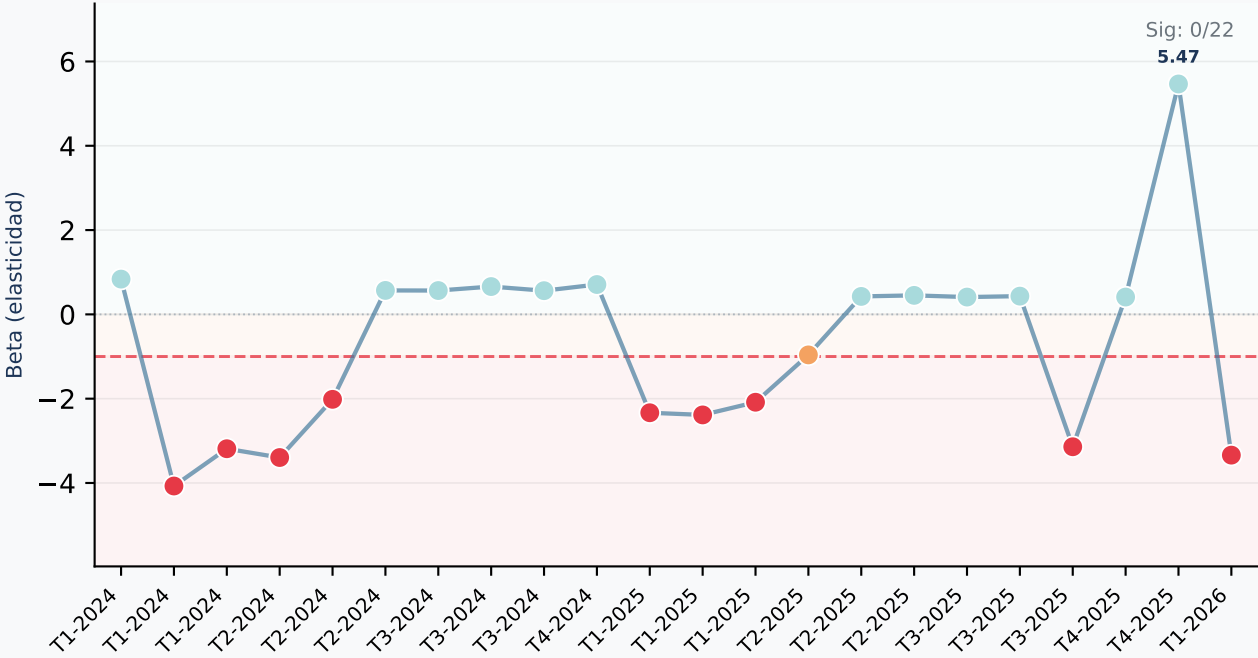


CARPETAS — Evolucion de Beta (Trimestral)
Cada punto = ventana de 3 mes(es). Verde = significativo (p<0.05)

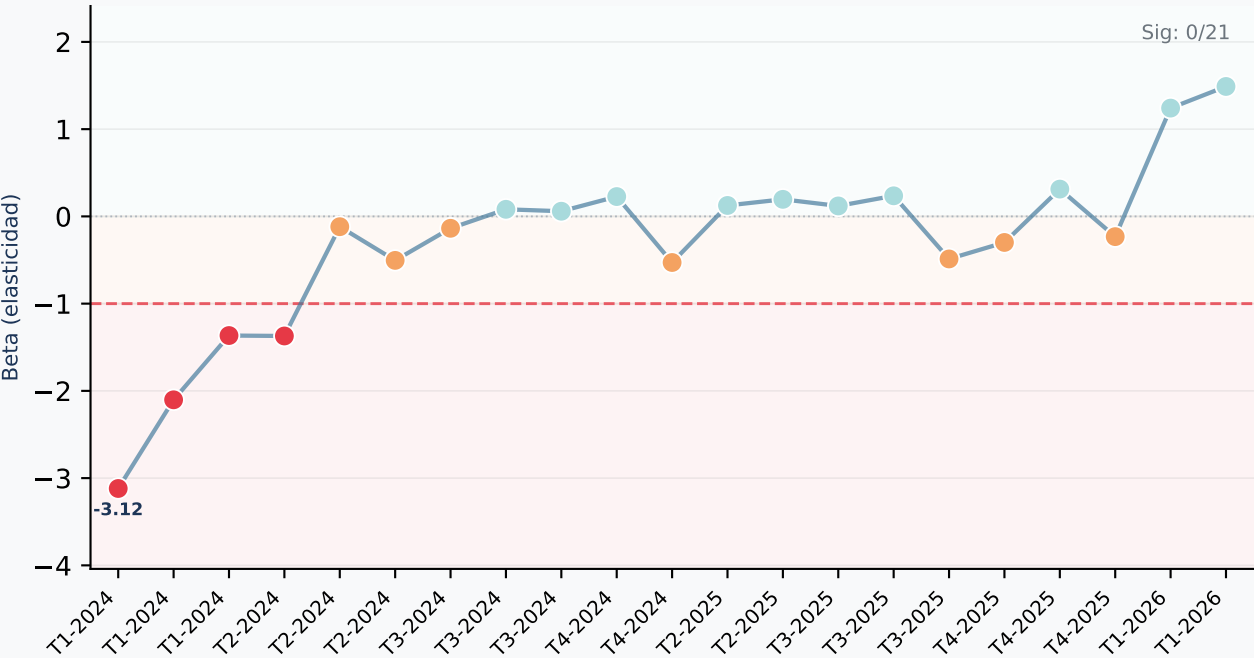
SEPARADORES VARIOS
SKU 50062074 | n=24 ventanas | CV precio=0.079



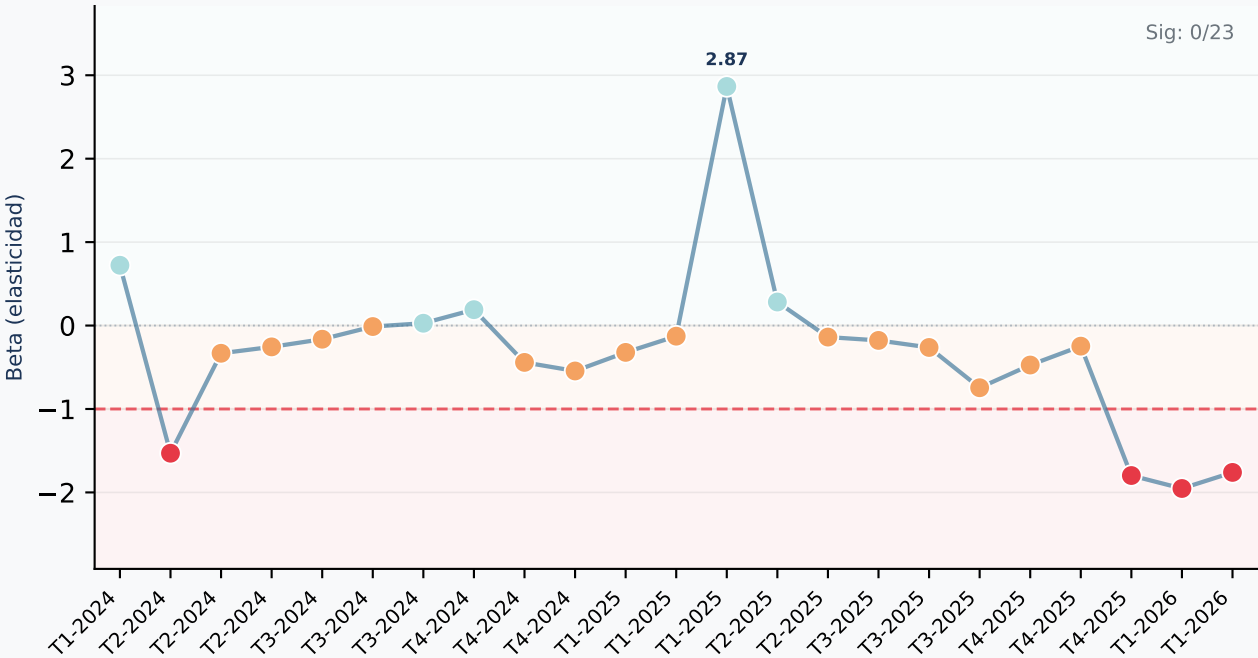
PROTECTORES PARA HOJAS
SKU 50062078 | n=22 ventanas | CV precio=0.085



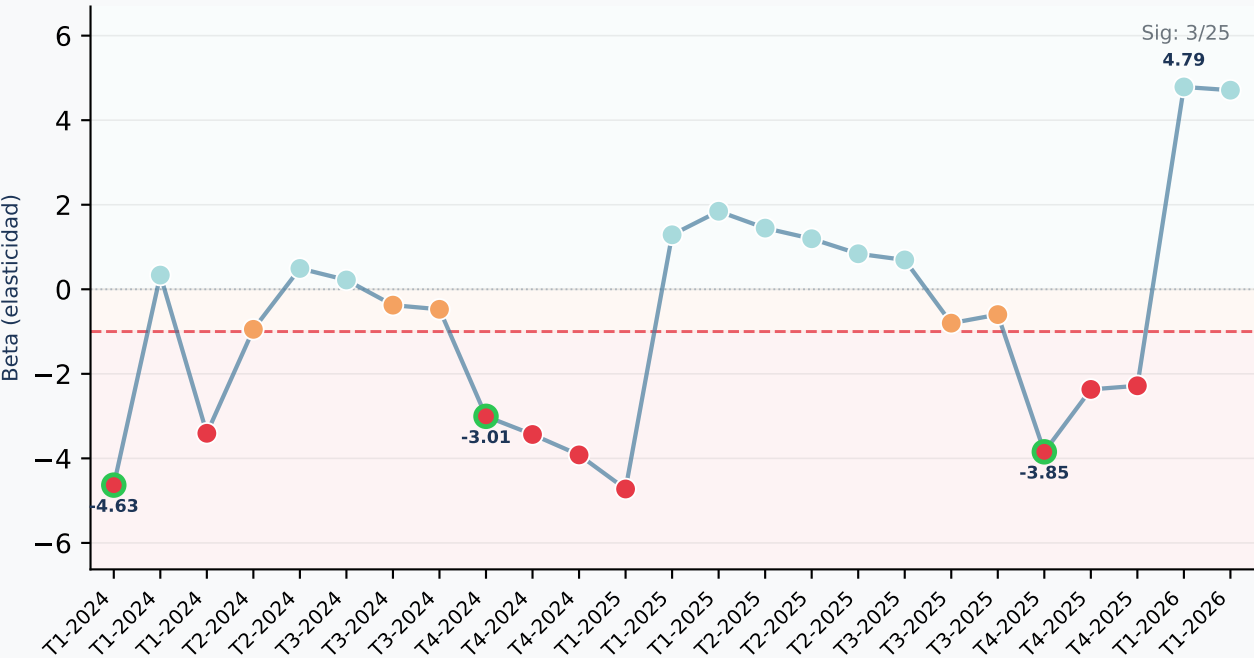
TABLAS DE PLASTICO
SKU 50062115 | n=21 ventanas | CV precio=0.089



PROTECTORES PARA HOJAS
SKU 50062270 | n=23 ventanas | CV precio=0.132

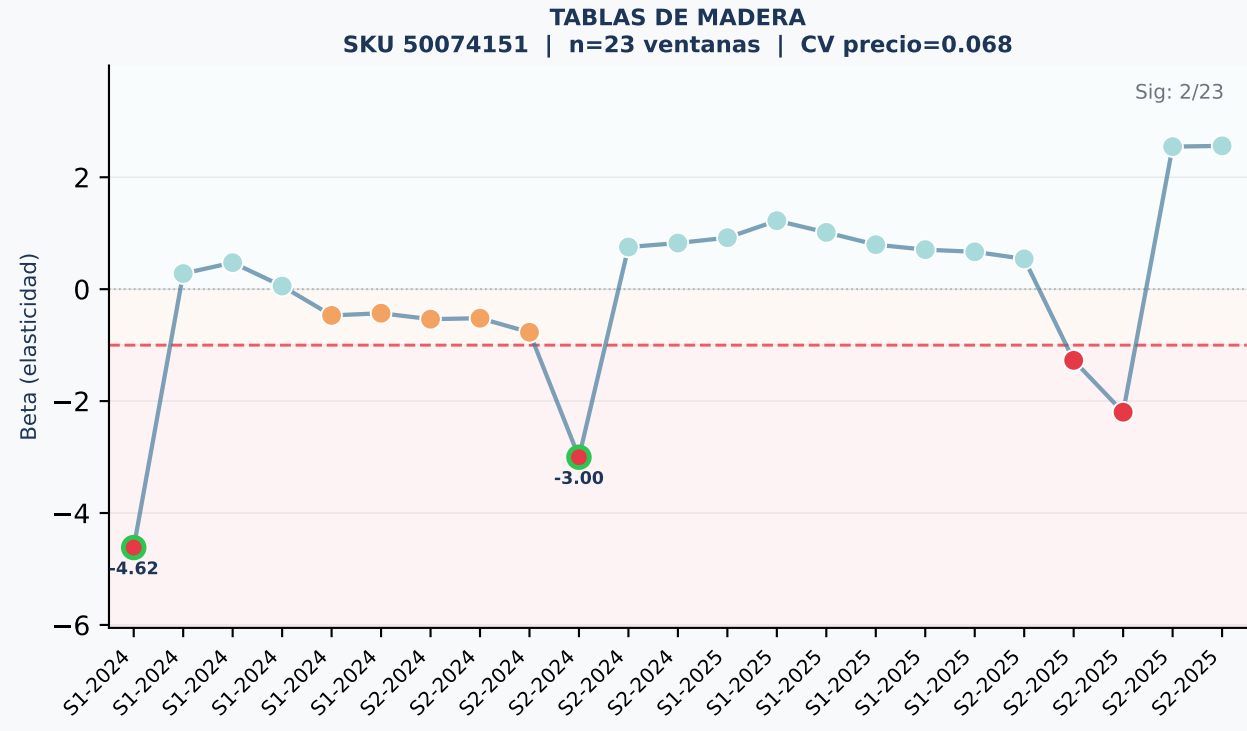
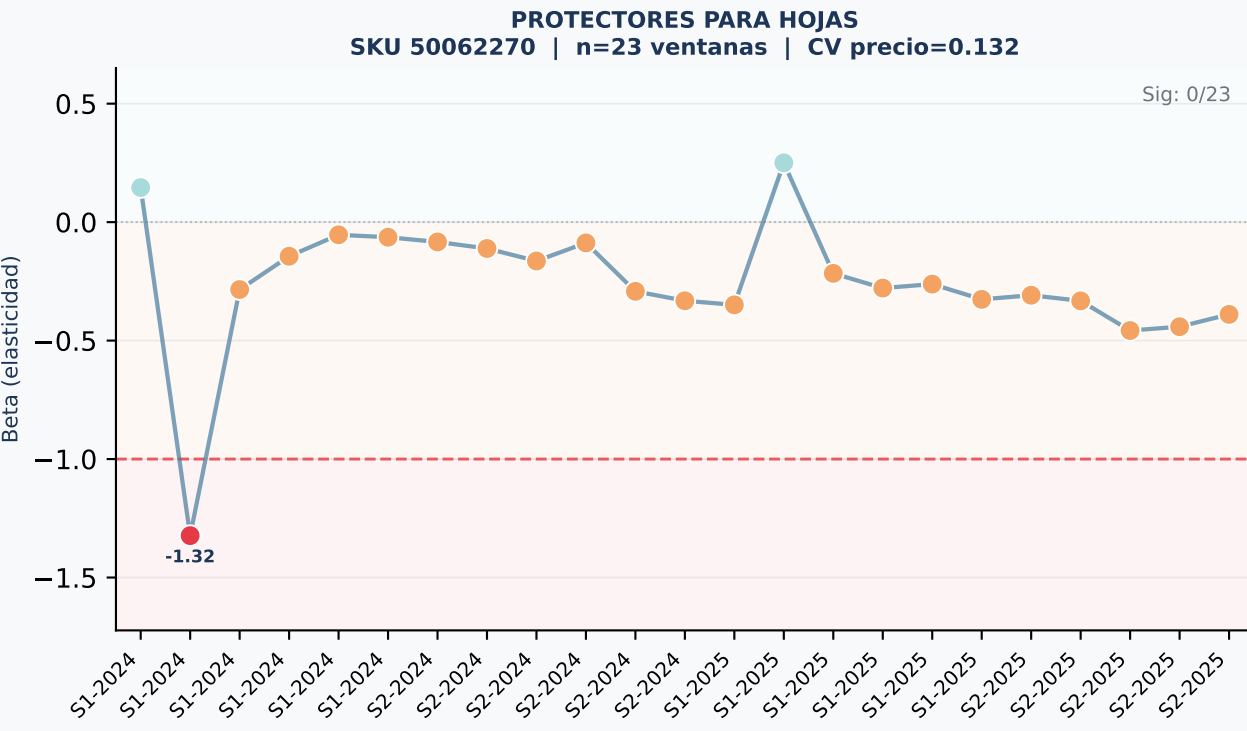
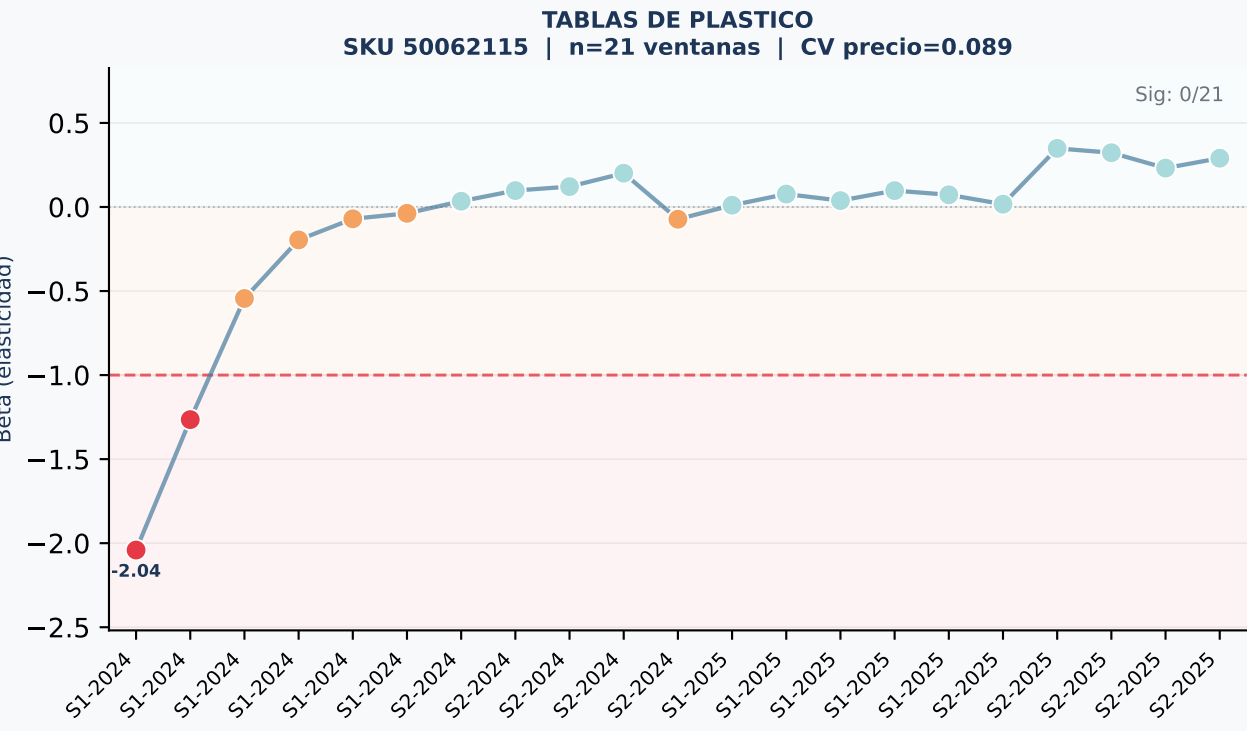
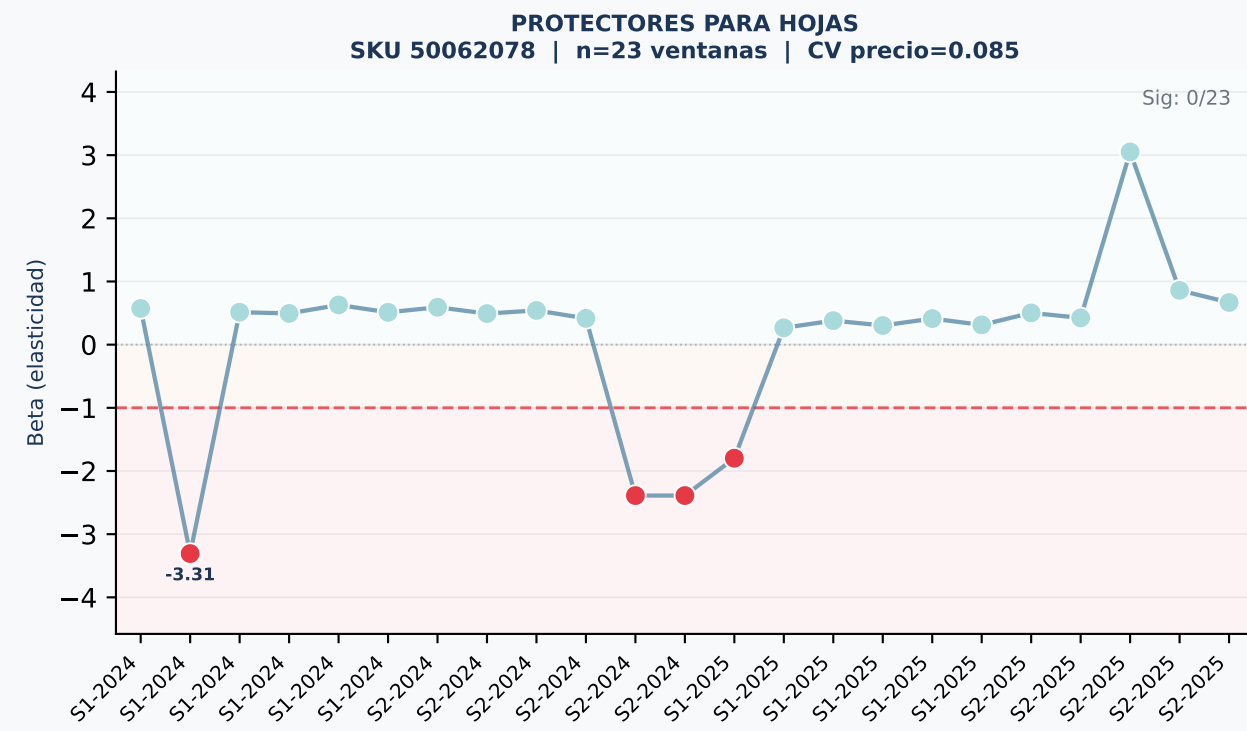
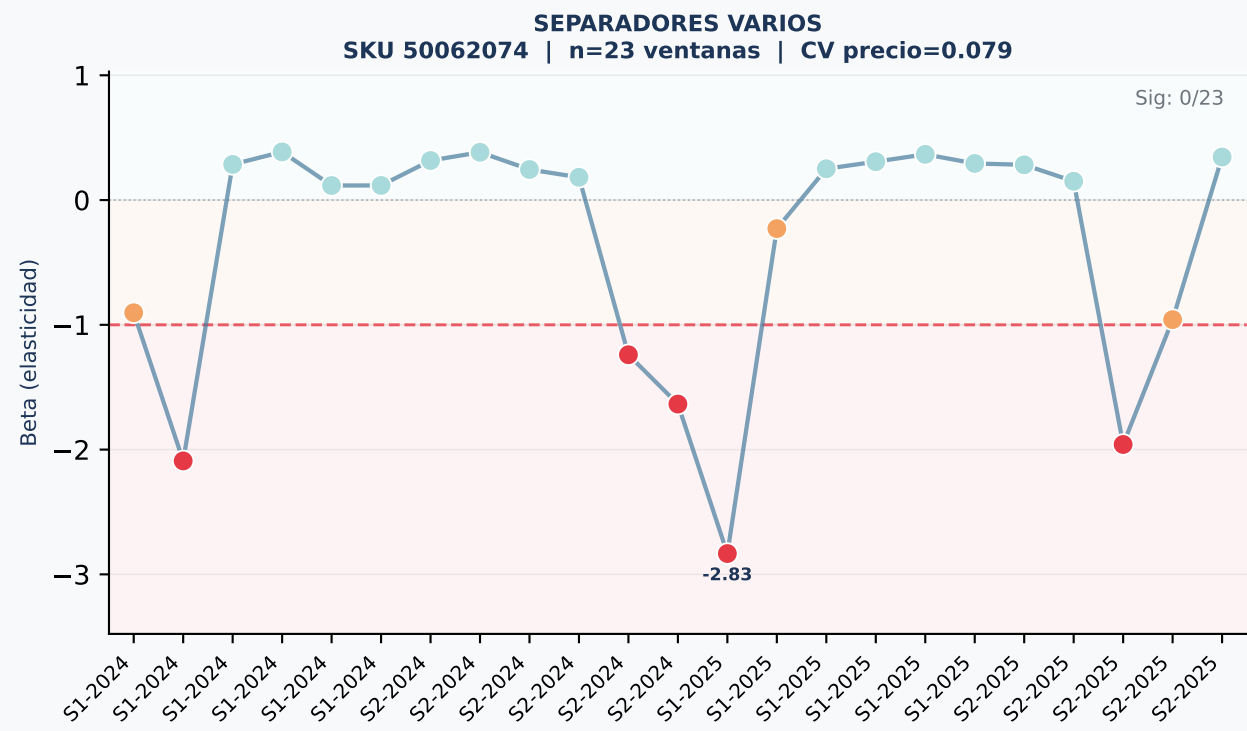


TABLAS DE MADERA
SKU 50074151 | n=25 ventanas | CV precio=0.068



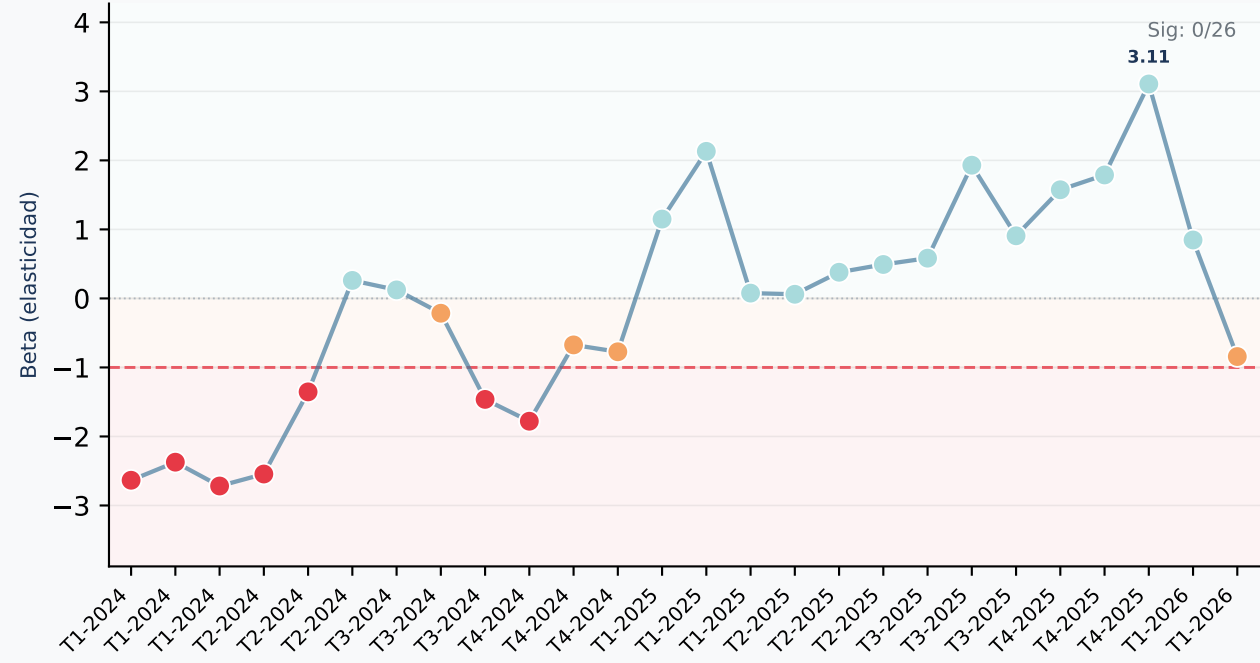
Elastico ($\beta < -1$) Inelastico ($-1 < \beta < 0$) Anomalo ($\beta \geq 0$) Punto significativo ($p < 0.05$)

CARPETAS — Evolucion de Beta (Semestral)
Cada punto = ventana de 6 mes(es). Verde = significativo (p<0.05)

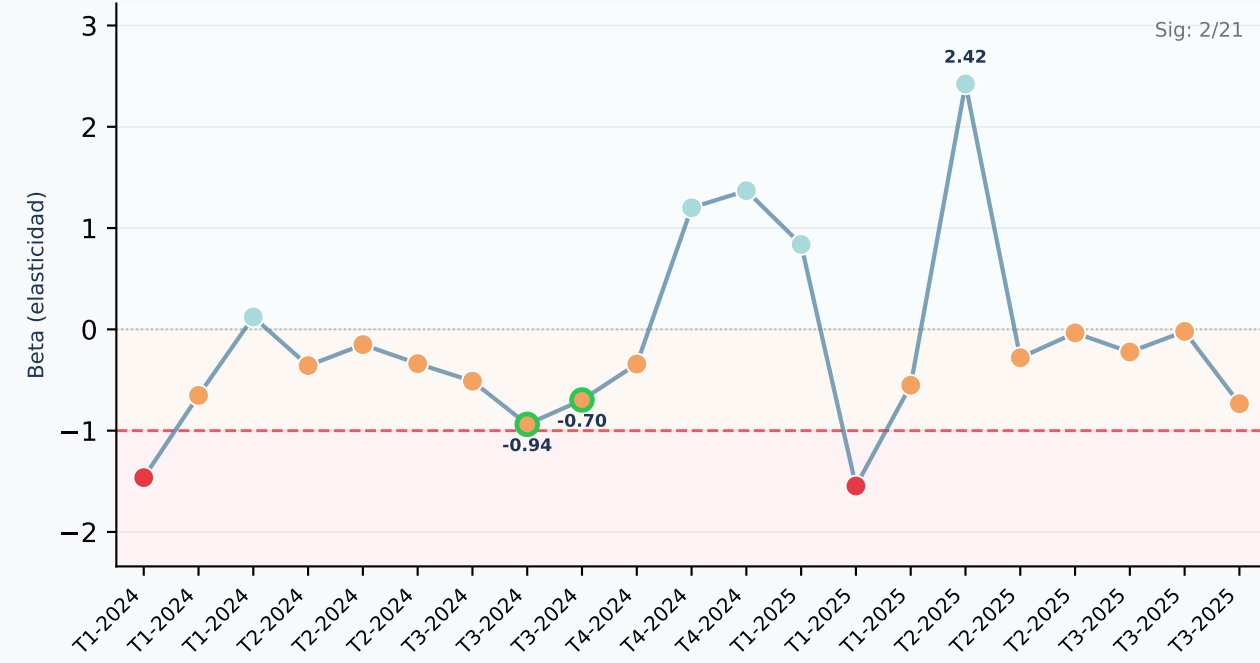


SUMINISTROS DE OFICINA — Evolucion de Beta (Trimestral)
Cada punto = ventana de 3 mes(es). Verde = significativo (p<0.05)

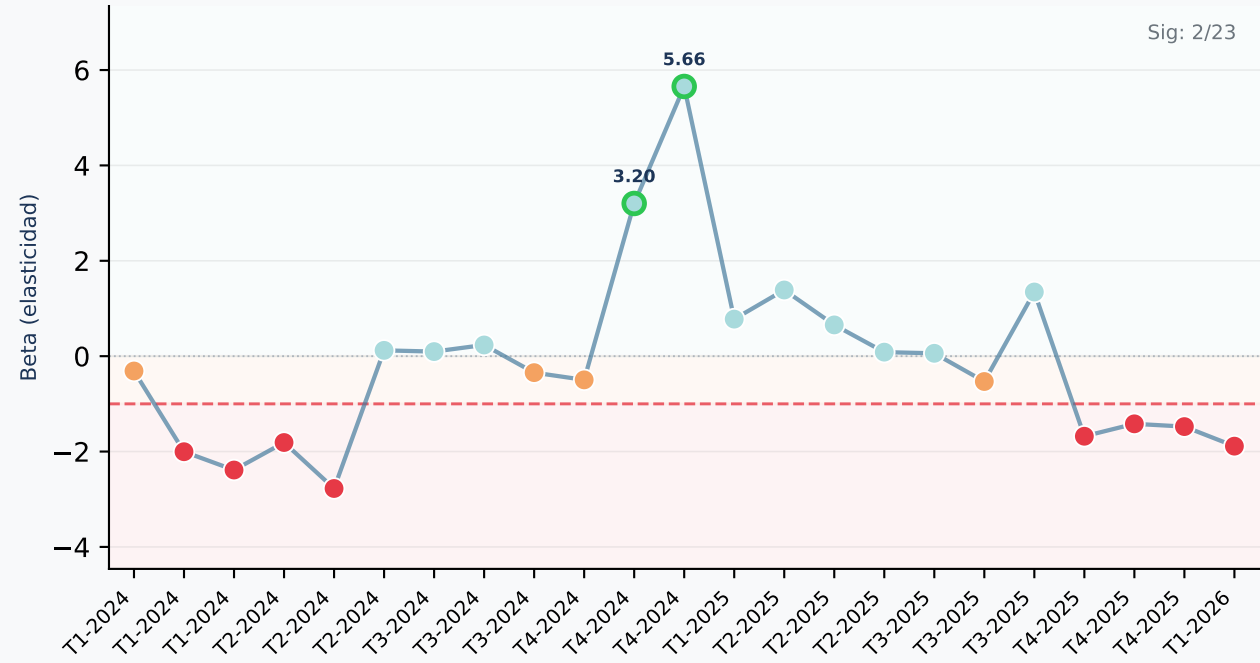
NOTAS ADHESIVAS
SKU 50061998 | n=26 ventanas | CV precio=0.260



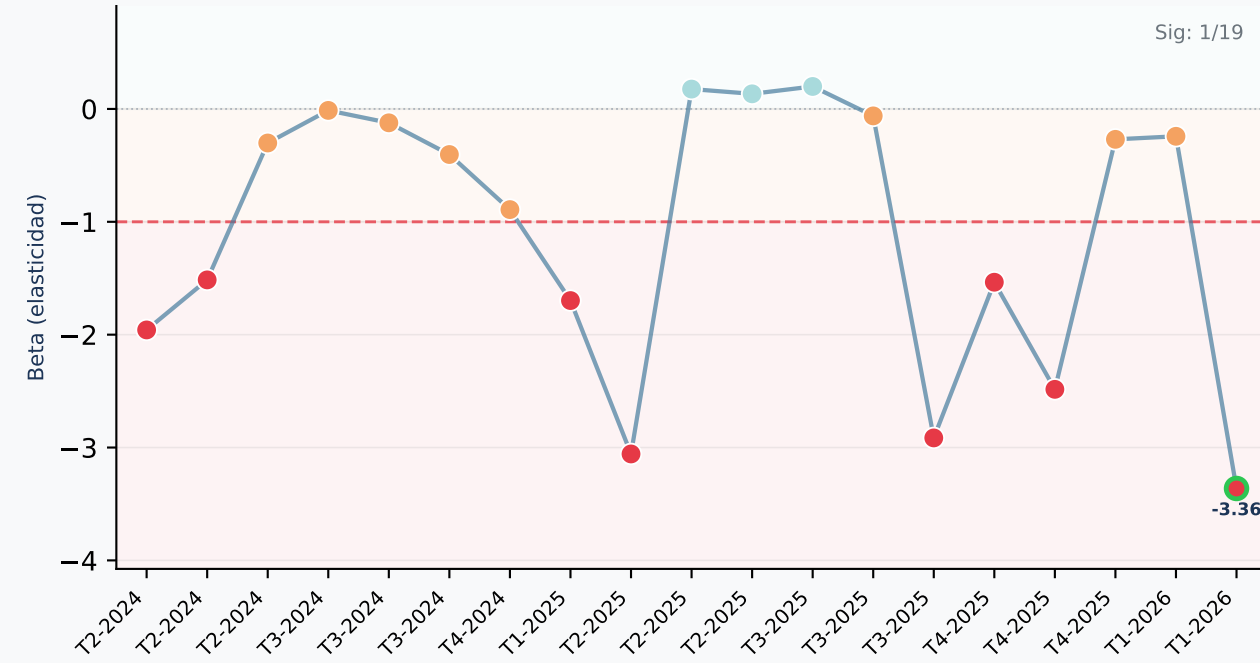
LAPIZ ADHESIVO
SKU 50062274 | n=21 ventanas | CV precio=0.234



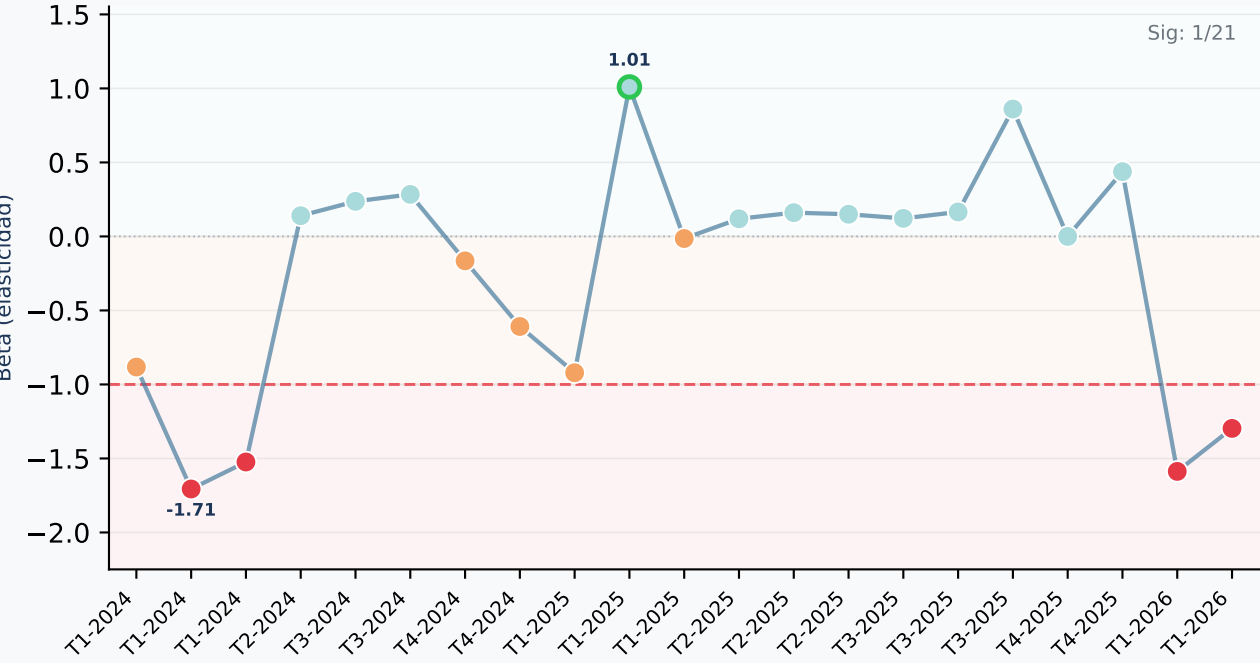
LAPIZ ADHESIVO
SKU 50062275 | n=23 ventanas | CV precio=0.103



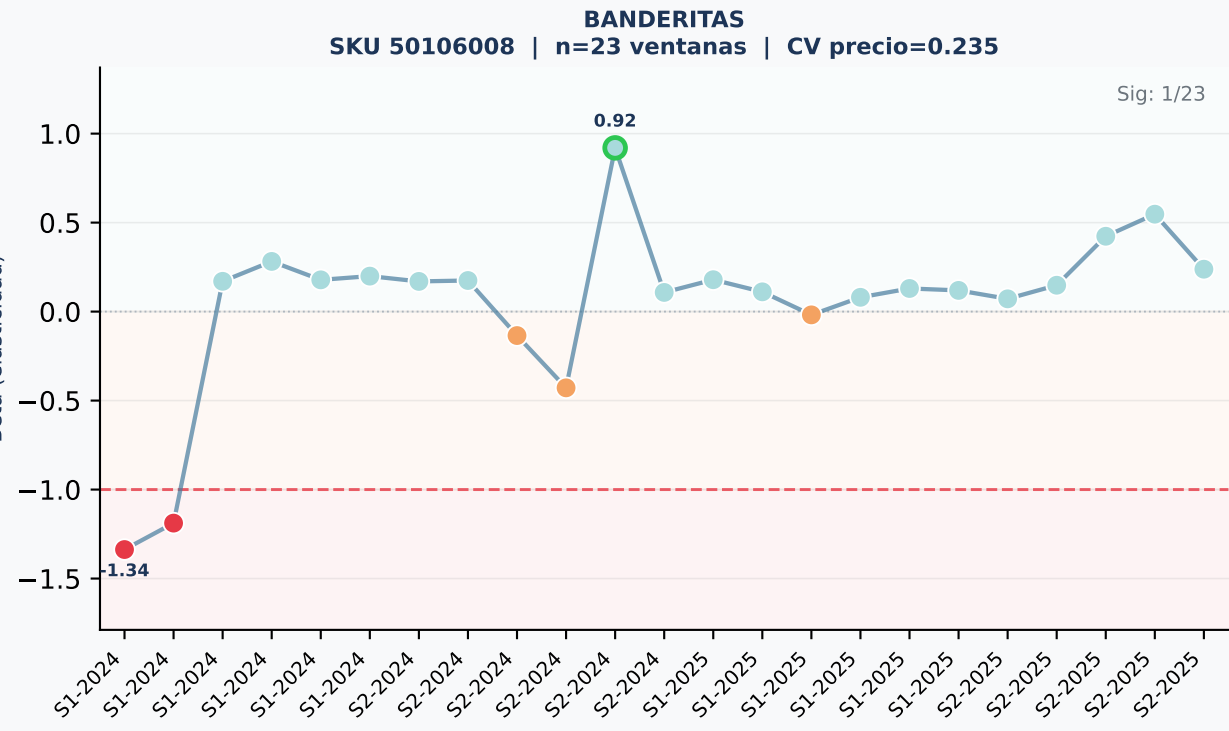
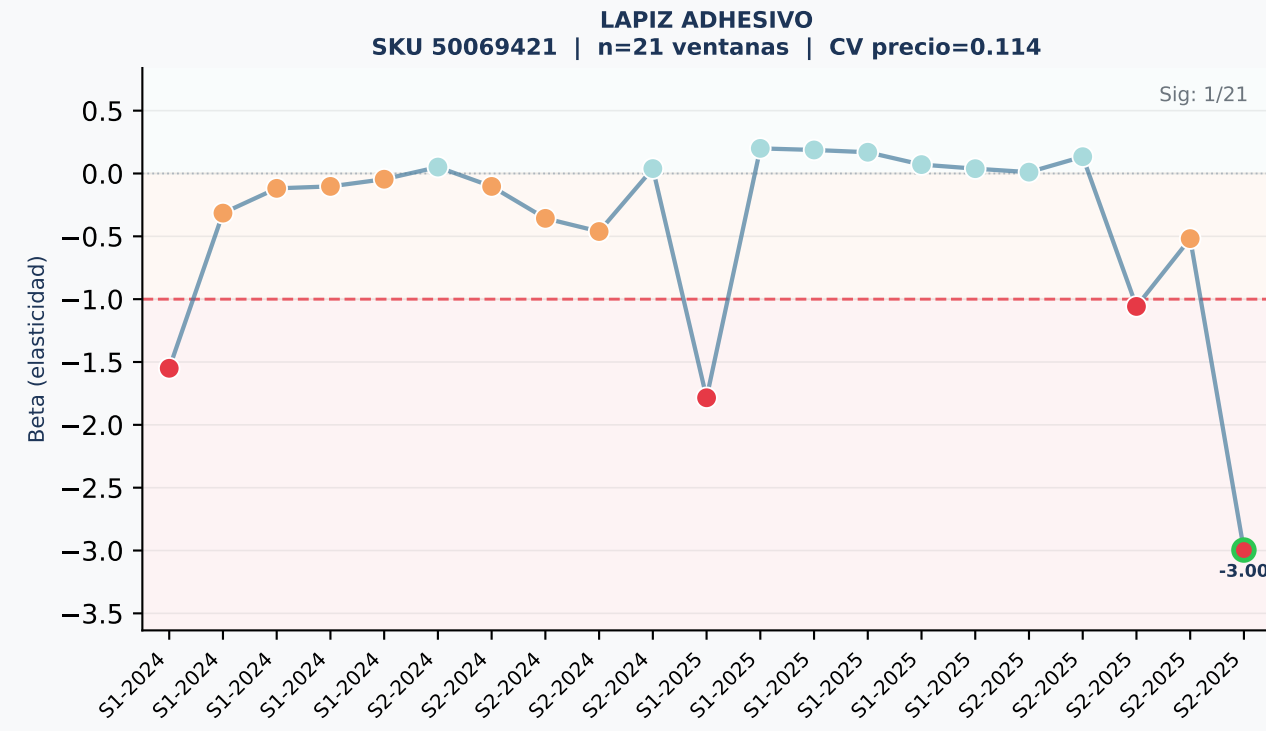
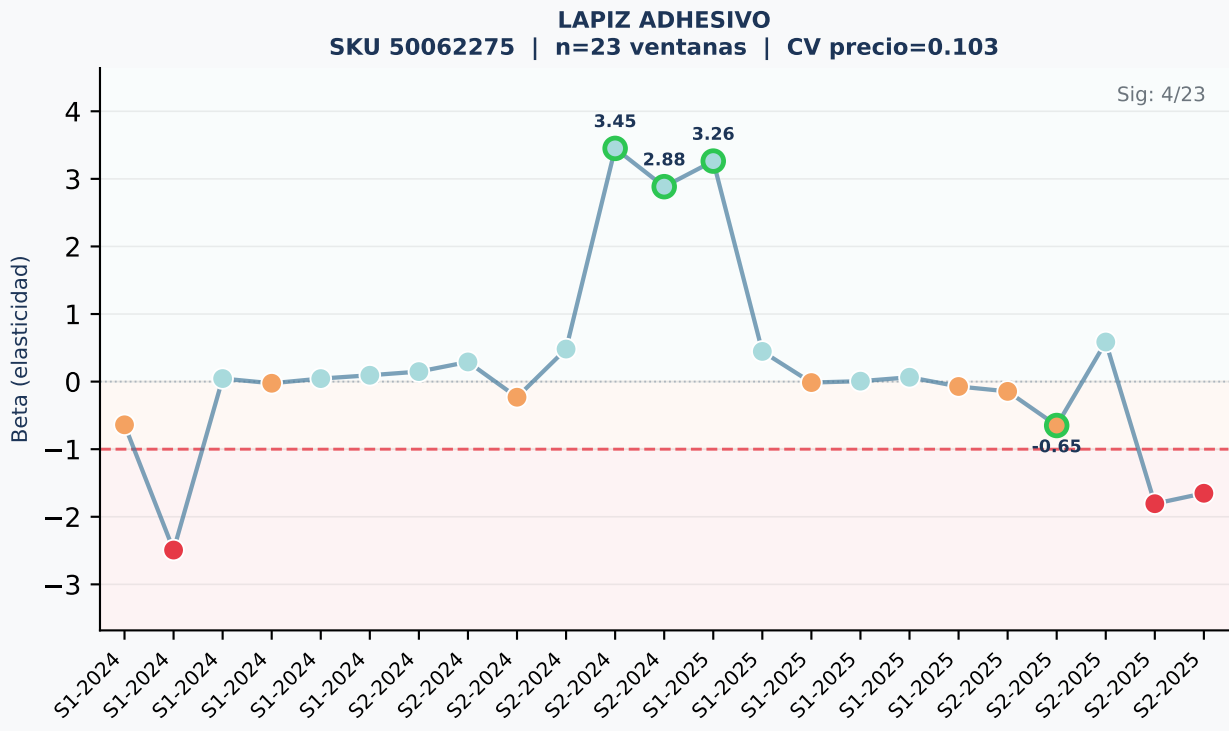
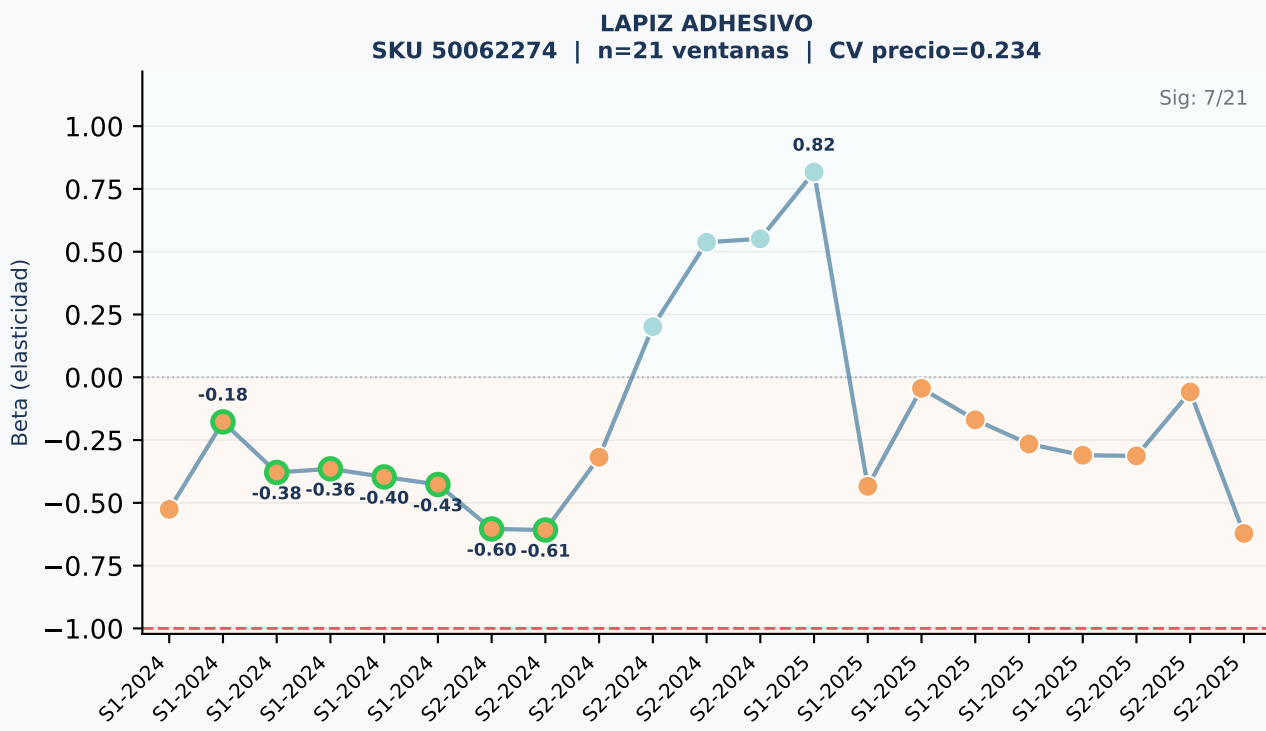
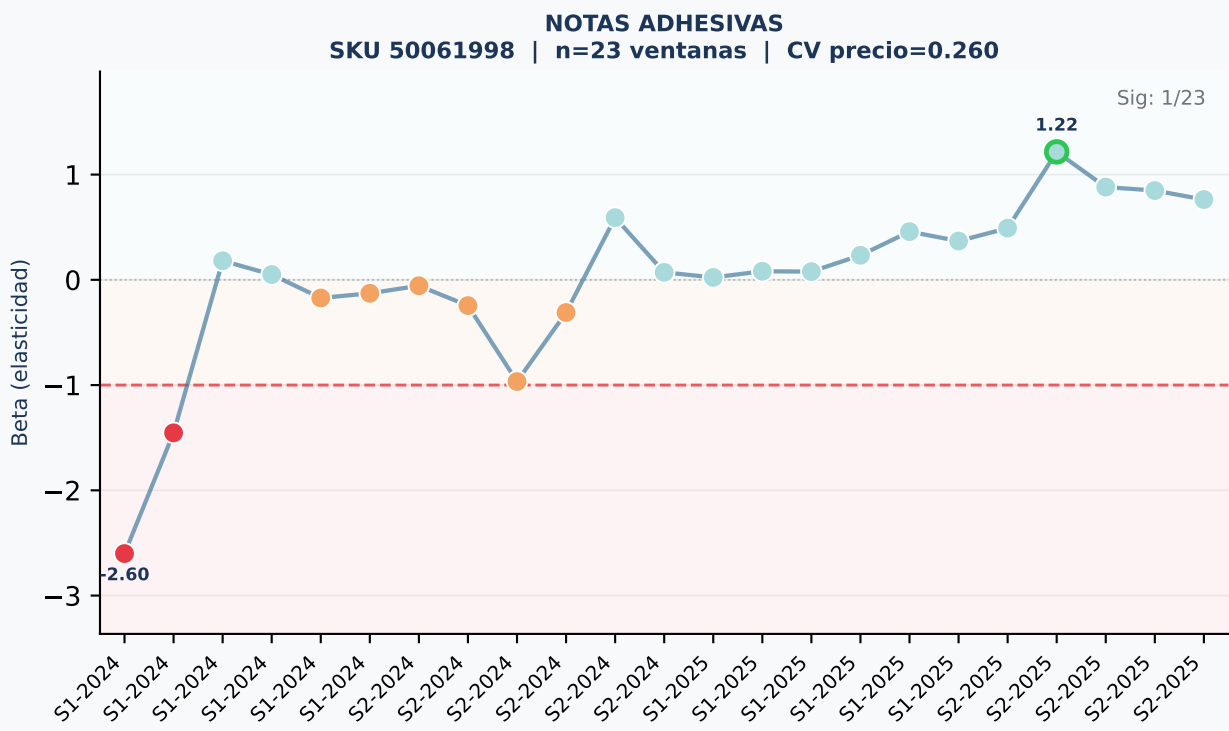
LAPIZ ADHESIVO
SKU 50069421 | n=19 ventanas | CV precio=0.114



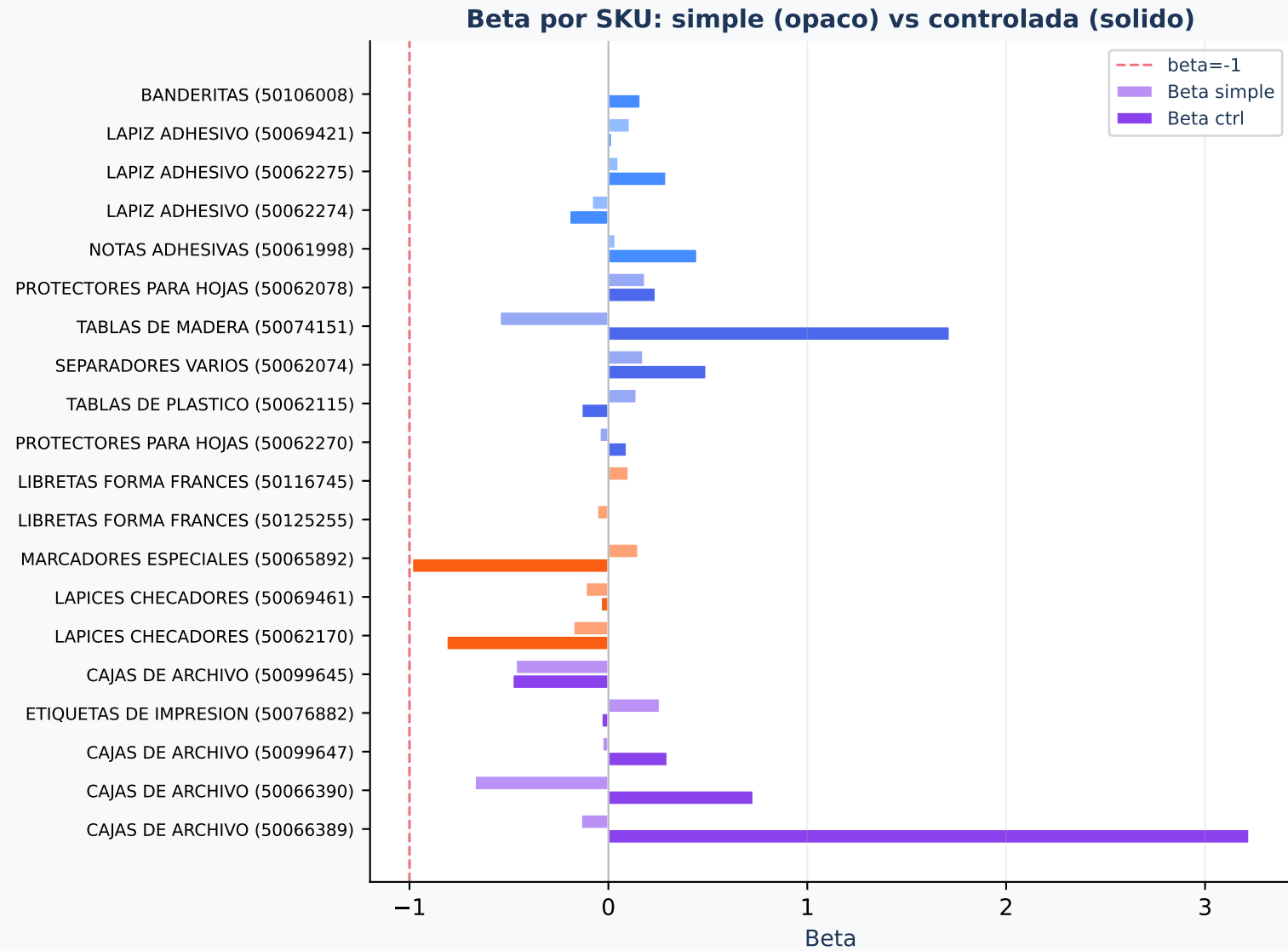
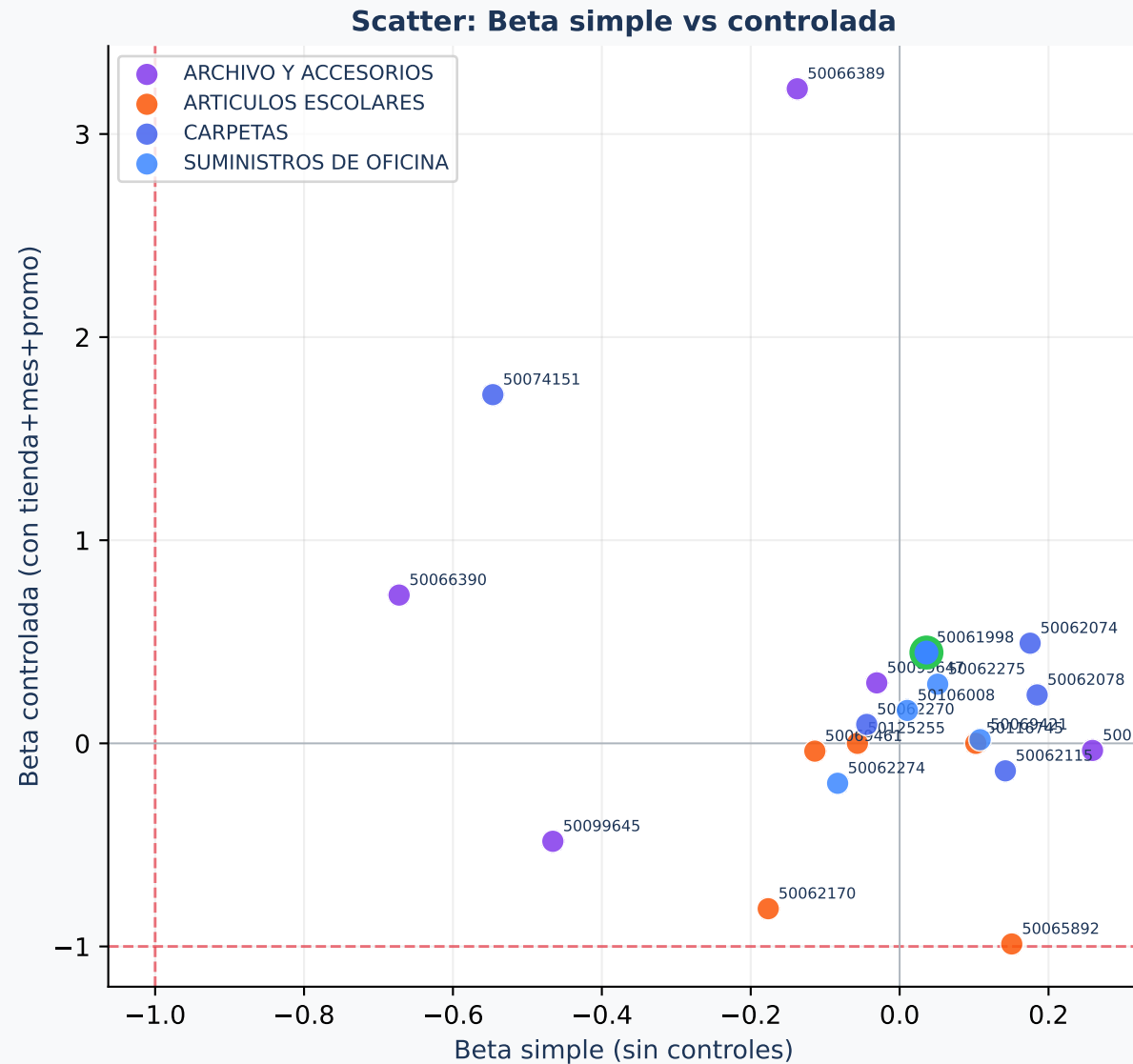
BANDERITAS
SKU 50106008 | n=21 ventanas | CV precio=0.235



SUMINISTROS DE OFICINA — Evolucion de Beta (Semestral)
Cada punto = ventana de 6 mes(es). Verde = significativo (p<0.05)

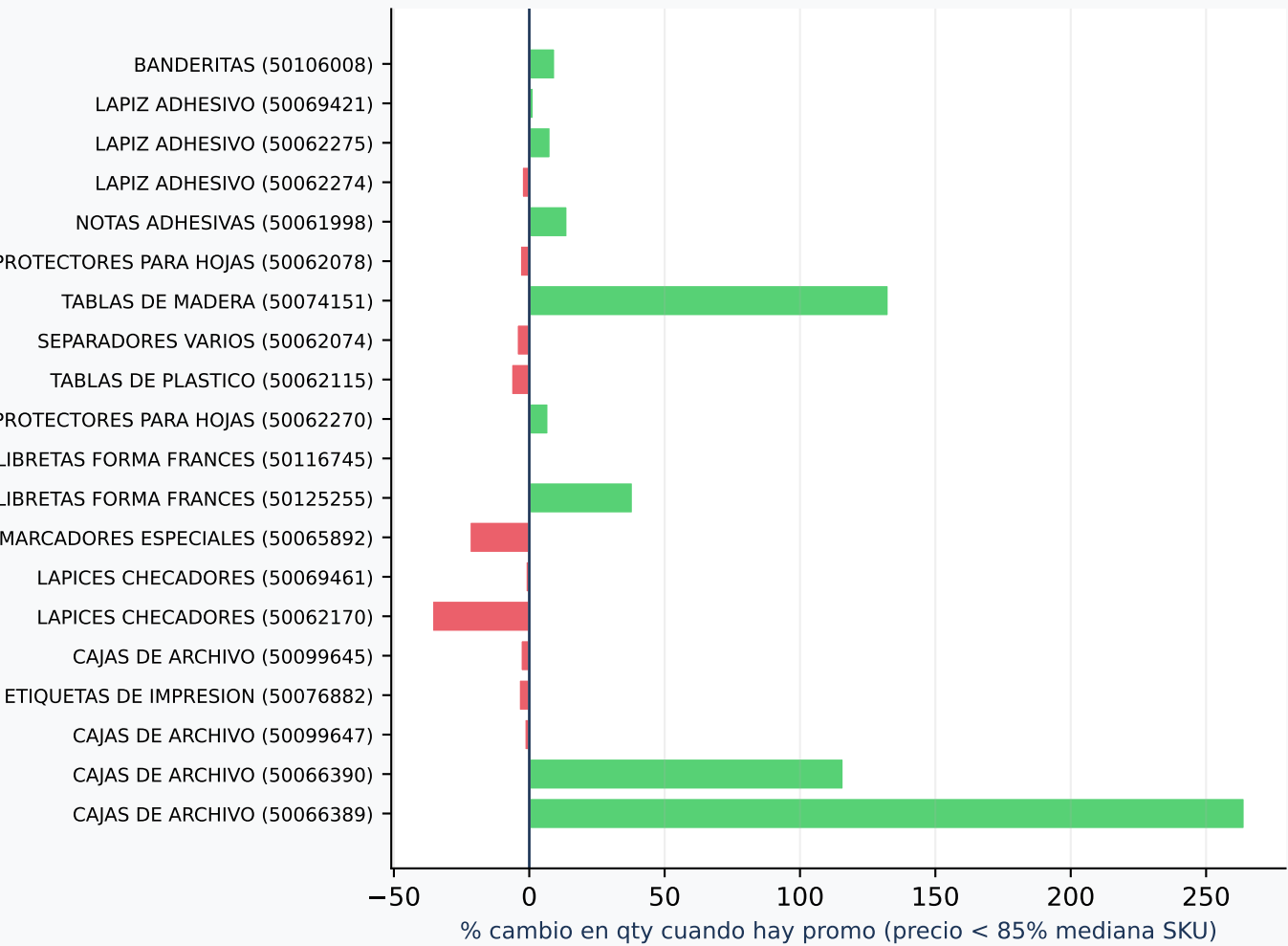


PARTE 2 - Comparacion de Beta: Modelo Simple vs Modelo con Controles (controles: tienda, mes, año, es_promo)

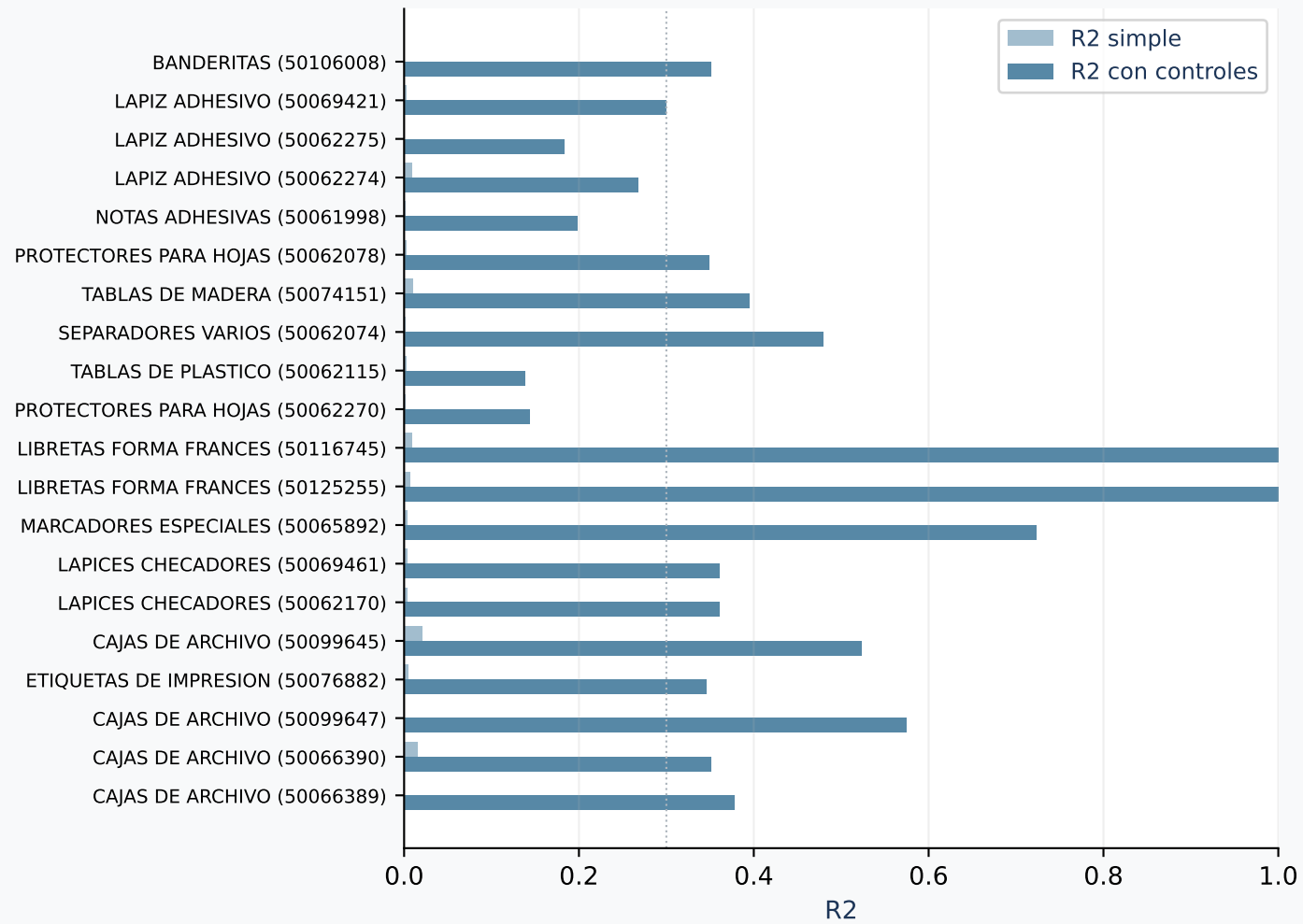


PARTE 2 - Efecto de Promocion y Poder Explicativo del Modelo

Efecto estimado de Promocion
 (// = significativo p<0.05)

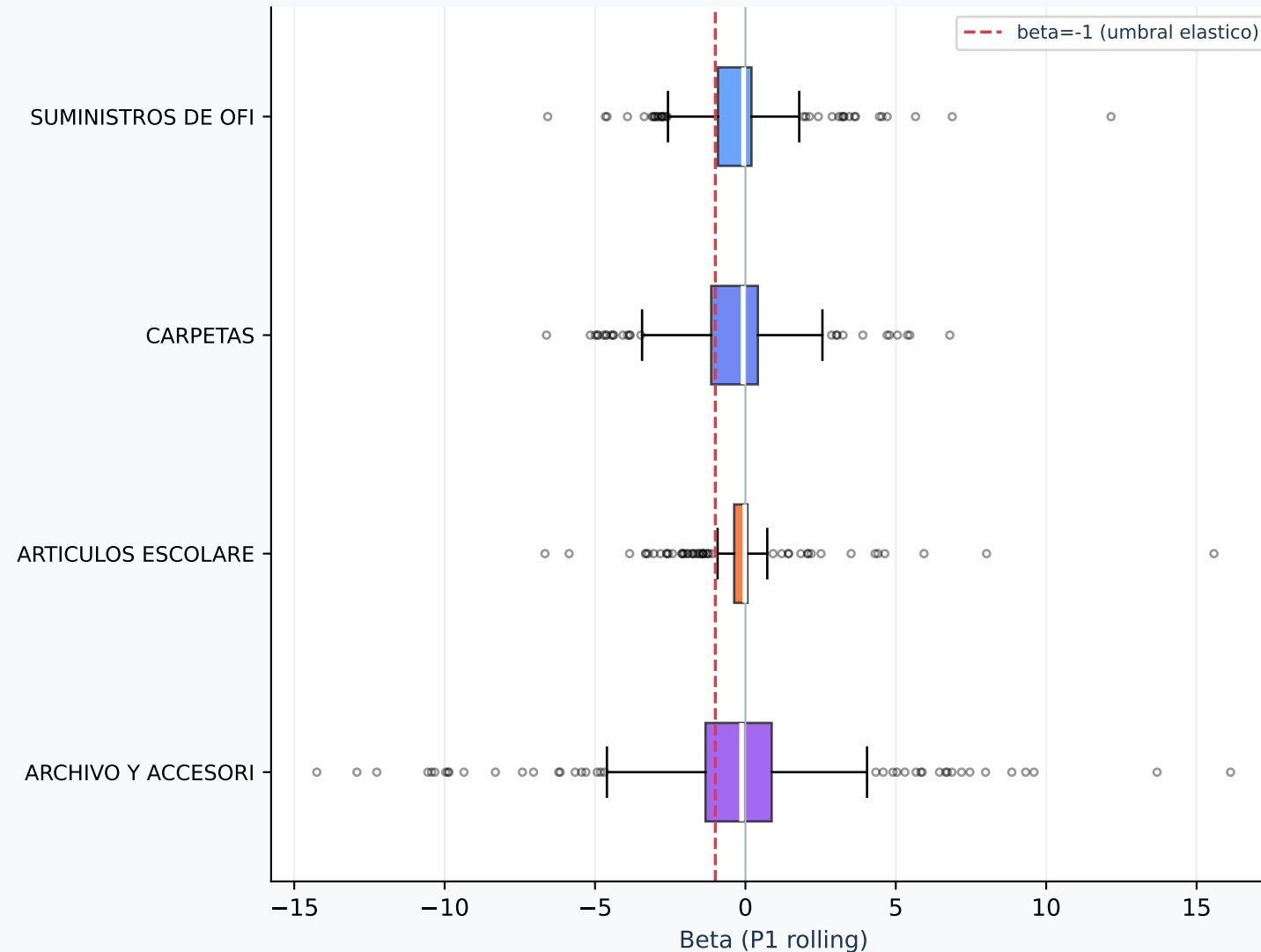


Poder explicativo: R2 simple vs R2 con controles
 (controles = tienda + mes + año + promo)



Resumen Final — Oportunidades de Promocion

Distribucion de Beta por Categoria
(todas las ventanas P1)



Confiabilidad: mas datos -> estimacion mas robusta

